



The short duration premium[☆]

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ABSTRACT

Stocks of firms with cash flows concentrated in the short term (i.e., short duration stocks) pay a large premium over long duration stocks. I empirically demonstrate that this premium (i) is long-lived and strong even among large firms, (ii) subsumes the value and profitability premia, and (iii) exposes investors to variation in expected returns, especially in times when the premium is high. These facts are consistent with an intertemporal model in which the marginal (long-term) investor dislikes expected return declines as they lead to lower expected wealth growth. The model also captures the positive relation between risk premia and bond duration.

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1. Introduction

Linking asset prices to properties of cash flows is at the heart of the asset pricing research agenda. But what cash flow characteristics matter, and how are they connected

to asset prices? Recent literature indicates that cash flow maturity is an important characteristic and is connected to asset prices through a downward sloping term structure of risk premia beyond the first few years (for a literature review, see [Binsbergen and Kojen, 2017](#)). The relatively low risk premia of long-term risky cash flows have important implications for the cross-section of stock returns, and exploring these implications is of fundamental importance for our understanding of asset prices.

This paper sheds light on this issue by studying equity portfolios sorted on cash flow duration, which effectively rank stocks based on whether firm value is concentrated in short- or long-term cash flows. I find a strong premium for short duration stocks and show that several of its properties are consistent with this premium existing in equilibrium because earning it requires exposure to reinvestment risk, which is undesirable from the perspective of long-term investors.

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To start, I develop and apply a novel firm-level measure of equity duration and find that from 1973 to 2018, short duration stocks paid a large premium (8.6% per year in value-weighted decile portfolios) relative to long duration stocks despite having lower market betas. This finding is consistent with the literature (e.g., [Lettau and Wachter, 2007](#); [Lettau and Wachter, 2011](#); [Weber, 2018](#)) and represents the starting point for my analysis.

I provide two new empirical facts about the short duration premium. First, the premium is long-lived (lasts for at least five years) and is strong even among large firms (market equity in the highest NYSE quintile). Second, after controlling for duration, the value and profitability premia disappear, and this result is stronger if value, profitability, and duration are all included in joint tests. Intuitively, if two firms have different growth opportunities, but the same profitability, the one with less growth opportunities (the value firm) will save/invest less and pay more of its resources over the short term, becoming a short duration company. Similarly, if two firms have different profitability, but similar growth opportunities, the more profitable one will have more resources to distribute to investors in the near future and become a short duration firm.

These new empirical facts suggest that the value and profitability premia are proxies for the short duration premium, which does not resemble an anomaly that concentrates in small stocks and quickly disappears. Consequently, understanding the fundamental driver of the short duration premium can help us better understand the cross-section of stock returns.

In this vein, I provide a novel explanation for why the short duration premium exists in financial markets. First, I build on the intertemporal capital asset pricing model (ICAPM) of [Campbell \(1993\)](#) to argue that long-term investors care about long-term wealth and thus price market risk (i.e., variation in current wealth) as well as reinvestment risk (i.e., variation in expected wealth growth). Second, I empirically show that investors can only earn the short duration premium by being exposed to substantial reinvestment risk. Third, I demonstrate that this reinvestment risk exposure is large enough to explain the short duration premium observed empirically with plausible estimates of relative risk aversion.

The reinvestment risk exposure of the short duration strategy originates from a simple mechanism. Decreases in expected wealth growth are induced by lower expected returns and thus are associated with increases in equity prices through lower discount rates. Since longer duration stocks are more sensitive to discount rates, their prices increase by more when expected wealth growth declines, inducing negative returns on the short duration strategy. Consequently, the short duration strategy is strongly exposed to reinvestment risk.

To complement the cross-sectional evidence, I also explore the time series dimension. I find that the short duration premium is substantially larger in periods in which earning the premium requires higher exposure to reinvestment risk. In particular, I argue that larger cross-sectional dispersion in equity duration induces higher reinvestment risk for the short duration strategy given the larger duration spread between sorted portfolios. Conse-

quently, we should observe a larger premium in periods with higher cross-sectional dispersion in equity duration if reinvestment risk is its main driver. Consistent with this hypothesis, I find that the short duration premium is much higher in periods with higher cross-sectional dispersion in equity duration (20% versus 3%). Moreover, the reinvestment risk of the short duration strategy is also much larger during these periods, which provides a time series link between the short duration premium and reinvestment risk.

It is natural to wonder whether capturing the short duration premium comes at the cost of producing higher risk premia for shorter duration bonds, which would be counterfactual. This is not the case; I find that the reinvestment risk mechanism helps generate risk premia that increase in bond duration for government and corporate bonds. This result is a consequence of the negative correlation between nominal interest rates and the equity premium ([Fama and Schwert, 1977](#); [Campbell, 1987](#); [Ferson, 1989](#); [Shanken, 1990](#); [Brennan, 1997](#); [Cederburg, 2019](#)). Specifically, interest rates rise and bond prices fall as expected equity returns decline, and this effect increases in bond duration, inducing higher reinvestment risk for longer duration government and corporate bonds.

In summary, I develop a novel measure of equity duration and use it to study the short duration premium extensively. I find that this premium is observed even among large firms, lasts for a long period, and dominates the value and profitability premia. Moreover, I show that several of the short duration premium properties are consistent with the hypothesis that this premium exists because earning it requires exposure to reinvestment risk, which is priced by long-term investors.

This paper complements the recent literature on the equity term structure (for a literature review, see [Binsbergen and Koijen, 2017](#)).¹ In particular, my findings strengthen the challenge that the downward sloping equity term structure poses to popular consumption-based asset pricing models such as the external habit model

¹ The equity term structure literature builds on [Brennan \(1998\)](#)'s insight that variation in dividend prices provides an important signal for changes in the fundamental value of cash flows. [Binsbergen, Brandt, and Koijen \(2012\)](#) use option prices and the put-call parity to show that claims on S&P 500 short-term dividends have higher average returns than the index itself, which is a long duration portfolio of dividend claims. [Binsbergen and Koijen \(2017\)](#) study dividend futures contracts and find similar results, while [Binsbergen et al. \(2013\)](#) emphasize the procyclicality of the term structure of hold-to-maturity risk premia, and [Gormsen \(2021\)](#) shows countercyclicality of the term structure of one-period holding returns. [Cejnek and Randl \(2020\)](#) and [Li and Wang \(2017\)](#) further explore the predictability power of dividend prices, and [Cejnek and Randl \(2016\)](#) relate the dividend term structure to an options-based downside risk factor. [Manley and Mueller-Glissmann \(2008\)](#) and [Mixon and Onur \(2017\)](#) provide valuable institutional details about the market for dividend derivatives. Some papers argue that frictions affecting dividend derivatives can explain the equity term structure ([Boguth et al., 2012](#); [Schulz, 2016](#); [Song, 2016](#); [Bansal et al., 2019](#)). However, [Binsbergen and Koijen \(2011, 2016\)](#) detail some important limitations of these friction-related mechanisms, and my results are immune to the frictions explored in these papers as my analysis is based on the cross-section of stock returns, providing additional support for the point in [Binsbergen and Koijen \(2011, 2016\)](#) that frictions cannot explain the equity term structure.

(Campbell and Cochrane, 1999), the long-run risks models (Bansal and Yaron, 2004), and the rare disasters model (Barro, 2006). New models need to explain not only the downward sloping equity term structure but also the closely connected short duration premium. One model that meets this challenge is the ICAPM as I demonstrate that it can explain the short duration premium and Gonçalves (2021) shows that it can explain the equity term structure. However, the ICAPM does not provide a microfoundation for the time-varying expected returns that create reinvestment risk. As such, providing new models that endogenize time-varying expected returns while being consistent with the equity term structure and the short duration premium is an important challenge for future research. Alternative frameworks that jointly explain the equity term structure and the short duration premium would also be valuable to the literature.

This paper is also connected to the literature linking cash flow duration to asset prices (Dechow, Sloan, and Soliman, 2004; Lettau and Wachter, 2007; Lettau and Wachter, 2011; Da, 2009; Chen, 2011; Weber, 2018; Chen and Li, 2018; Gormsen and Lazarus, 2019). My measure of equity duration (*Dur*) is most closely related to the equity duration measure in Dechow, Sloan, and Soliman (2004) (*DSS Dur*). In Section 5, I provide a comparison between *Dur* and *DSS Dur* to demonstrate that despite *DSS Dur* being an important first step, *Dur* substantially improves upon it on theoretical and empirical grounds.

Weber (2018) is the closest paper to mine in terms of the objective of better understanding what drives the short duration premium. He studies duration-sorted portfolios based on *DSS Dur* and argues that market participants are overly optimistic about the prospects of long duration companies, inducing overvaluation for these companies, which leads to poor returns. While I provide a direct link between equity duration and reinvestment risk both in the cross-section and in the time series, Weber (2018) links equity duration to mispricing (also on both dimensions). In Section 5, I discuss the relation between the two papers and argue that reinvestment risk and mispricing are complementary channels that help capture the short duration premium. Specifically, reinvestment risk justifies the existence of a short duration premium in the absence of mispricing, while mispricing generates a larger short duration premium among stocks that are more subject to mispricing.

The result that the value premium disappears after controlling for duration is predicted by the prominent theoretical framework presented in Lettau and Wachter (2007, 2011). However, I empirically demonstrate the underlying cash flow risk mechanism proposed in these papers does not perform well empirically in explaining the short duration premium I observe in the data, especially its time variation. Of course, one of the main appeals of the framework in Lettau and Wachter (2007, 2011) is that it combines elements of structural models with aspects of reduced-form models and thus is flexible enough to incorporate new effects. As such, exploring adjustments to this framework that capture different asset pricing phenomena (including the time variation in the short duration premium) remains a fruitful research agenda.

My findings are also related to the ICAPM literature.² The paper closest to mine in this literature is Campbell and Vuolteenaho (2004). They empirically show that the value premium can be explained by the ICAPM of Campbell (1993). The cross-sectional link between reinvestment risk and equity duration I find is theoretically expected once we combine the empirical results in Campbell and Vuolteenaho (2004) with the theoretical argument that we should observe a strong correlation between equity duration and valuation ratios (Lettau and Wachter, 2007; Lettau and Wachter, 2011). My paper adds to these previous papers by exploring this cross-sectional link empirically and by showing that, after controlling for duration, the value and profitability premia both disappear. I also add to the previous literature by providing evidence that connects the time series variation in the short duration premium to reinvestment risk. Finally, my empirical results build a direct bridge between equity duration and the equity term structure as reinvestment risk can explain both phenomena.

Two contemporaneous and independent working papers (Chen and Li, 2018; Gormsen and Lazarus, 2019) also demonstrate the link between duration and other cross-sectional asset pricing anomalies (such as value and profitability). Despite this similarity, our papers are fundamentally different. While I focus on understanding the intrinsic driver of the short duration premium, Chen and Li (2018) and Gormsen and Lazarus (2019) take the short duration premium as given (as a risk factor) to summarize the cross-section of stock returns.

The rest of the paper is organized as follows. Section 2 provides details on my equity duration measure and the empirical design. Sections 3 and 4 explore the short duration premium and its relation to reinvestment risk in the context of the ICAPM. In turn, Section 5 studies other equity duration measures, and Section 6 concludes. The Internet Appendix contains technical derivations, empirical details, and supplementary results.

2. Empirical design

This section details the definition and measurement of equity duration and provides some basic information on the sample construction, including the duration-sorted portfolios later used to study the implications of equity duration to the cross-section of stock returns. Technical derivations are provided in the Internet Appendix.

2.1. Equity duration

I define equity duration analogously to the common concept of bond duration (Macaulay, 1938) and demonstrate how it can be solved for under the assumption that

² The ICAPM was first proposed by Merton (1973), and a closely related discrete version was developed by Campbell (1993), which is the framework I build on. Given the long history of the ICAPM, many papers empirically tests and/or further develop the model. Some examples are Campbell (1996), Ferson and Harvey (1999), Brennan, Wang, and Xia (2004), Campbell and Vuolteenaho (2004), Petkova (2006), Campbell, Polk, and Vuolteenaho (2009), Bali and Engle (2010), Campbell et al. (2018), and Cederburg (2019).

(log) profitability and growth evolve linearly. Throughout the text, t represents a time index and j a firm index.

2.1.1. Definition

Duration is a concept often used to capture the average maturity (in years) of cash flows associated with a given investment. In general, duration is defined as

$$Dur_{j,t} = \sum_{h=1}^{\infty} w_{j,t}^{(h)} \cdot h, \quad (1)$$

where $w_{j,t}^{(h)} = (\mathbb{E}_t[CF_{j,t+h}] \cdot e^{-h \cdot dr_{j,t}}) / V_{j,t}$, with V representing the investment value, CF its cash flows, and dr its discount rate.

Intuitively, $w_{j,t}^{(h)}$ tells us the fraction of the investment value that is due to the cash flow maturing in h years. As such, w s are weights ($\sum_{h=1}^{\infty} w_{j,t}^{(h)} = 1$) and $Dur_{j,t}$ is a weighted average of cash flow maturities with weights depending on how important the given cash flow is to the current investment value. A low (or short) duration tells us that most of the investment value is due to short-term cash flows and the opposite is true for a high (or long) duration.

For US Treasury bonds, calculating duration is trivial since $CF_{j,t+h}$ is known as of t so that $dr_{j,t}$ is simply the (log) bond yield, which is also observable. For investments with risky cash flows, however, estimating $\mathbb{E}_t[CF_{j,t+h}]$ and $dr_{j,t}$ is a challenge.

To define equity duration, I treat firms' payout (dividends + repurchases - issuances), PO , as cash flows to equity investors so that V is the firm's market equity, ME , and $w_{j,t}^{(h)} = (\mathbb{E}_t[PO_{j,t+h}] \cdot e^{-h \cdot dr_{j,t}}) / ME_{j,t}$. Consequently, $dr_{j,t}$ can be defined as the discount rate that satisfies $\sum_{h=1}^{\infty} w_{j,t}^{(h)} = 1$ or, equivalently, the $dr_{j,t}$ that solves the valuation equation:

$$ME_{j,t} = \sum_{h=1}^{\infty} \mathbb{E}_t[PO_{j,t+h}] \cdot e^{-h \cdot dr_{j,t}}. \quad (2)$$

Eqs. (1) and (2) together with $V = ME$ and $CF = PO$ provide the full definition of equity duration.³

2.1.2. Measurement

The entire challenge in measuring equity duration is on estimating $\mathbb{E}_t[PO_{j,t+h}]$. To do so, let clean surplus earnings be $CSE_{j,t} = PO_{j,t} + \Delta BE_{j,t}$, with BE representing book equity and Δ the difference operator. Then, substituting the CSE definition into $\mathbb{E}_t[PO_{j,t+h}]$ gives

$$\begin{aligned} \frac{\mathbb{E}_t[PO_{j,t+h}]}{BE_{j,t}} &= \mathbb{E}_t \left[\left(1 + \frac{CSE_{j,t+h}}{BE_{j,t+h-1}} - \frac{BE_{j,t+h}}{BE_{j,t+h-1}} \right) \cdot \prod_{\tau=1}^{h-1} \frac{BE_{j,t+\tau}}{BE_{j,t+\tau-1}} \right] \\ &= \mathbb{E}_t \left[\left(e^{CSP_{j,t+h} - BE_{j,t+h}} - 1 \right) \cdot e^{\sum_{\tau=1}^h BE_{j,t+\tau}} \right], \end{aligned} \quad (3)$$

where the second equality follows from the definitions $CSP_{j,t} = \log(1 + CSE_{j,t}/BE_{j,t-1})$ and $BE_{j,t} = \log(BE_{j,t}/BE_{j,t-1})$.

³ In Section 5, I explore two alternative equity duration measures that do not require the $dr_{j,t}$ estimation and find results consistent with the ones obtained from my baseline equity duration measure.

To estimate $\mathbb{E}_t[PO_{j,t+h}]/BE_{j,t}$ at the firm-level, I follow Vuolteenaho (2002) and Campbell, Polk, and Vuolteenaho (2009) and assume $s_{j,t}$ is a vector of firm level characteristics (including a constant, $CSP_{j,t}$, and $BE_{j,t}$) that follows a vector autoregressive (VAR) system of order one:

$$s_{j,t} = \Gamma s_{j,t-1} + u_{j,t}, \quad (4)$$

where $u_{j,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \Sigma)$ with arbitrary cross-sectional covariance structure.

Using the VAR system in Eq. (4), we have (see the Internet Appendix for details)

$$\begin{aligned} \frac{\mathbb{E}_t[PO_{j,t+h}]}{BE_{j,t}} &= \left[e^{(\mathbf{1}_{CSP} - \mathbf{1}_{BE})' \Gamma^h s_{j,t} + v_1(h)} - 1 \right] \\ &\quad \cdot e^{\mathbf{1}_{BE}' (\sum_{\tau=1}^h \Gamma^\tau) \cdot s_{j,t} + h \cdot v_2(h)}, \end{aligned} \quad (5)$$

where $\mathbf{1}_x$ is a selector vector such that $\mathbf{1}_x' s_{j,t} = x_{j,t}$ and $v_i(h)$ are parameters that depend on Γ , Σ , and h but not on the state vector so that they do not vary cross-sectionally.

Substituting Eq. (5) into Eq. (2), we can find the firm discount rate as the $dr_{j,t}$ that solves

$$\begin{aligned} ME_{j,t}/BE_{j,t} &= \sum_{h=1}^{\infty} \left[e^{(\mathbf{1}_{CSP} - \mathbf{1}_{BE})' \Gamma^h s_{j,t} + v_1(h)} - 1 \right] \\ &\quad \cdot e^{\mathbf{1}_{BE}' (\sum_{\tau=1}^h \Gamma^\tau) \cdot s_{j,t} - h \cdot dr_{j,t} + h \cdot v_2(h)} \end{aligned} \quad (6)$$

and calculate duration by rewriting Eq. (1) as

$$\begin{aligned} Dur_{j,t} &= \frac{BE_{j,t}}{ME_{j,t}} \cdot \sum_{h=1}^{\infty} h \cdot \left[e^{(\mathbf{1}_{CSP} - \mathbf{1}_{BE})' \Gamma^h s_{j,t} + v_1(h)} - 1 \right] \\ &\quad \cdot e^{\mathbf{1}_{BE}' (\sum_{\tau=1}^h \Gamma^\tau) \cdot s_{j,t} + h \cdot v_2(h) - h \cdot dr_{j,t}}. \end{aligned} \quad (7)$$

I use a root-finding algorithm for each firm-year observation separately on Eq. (6) to solve for $dr_{j,t}$ given $s_{j,t}$ and the VAR parameter estimates for Γ and Σ .⁴ I then use Eq. (7) to calculate duration for the given firm-year observation. As a consequence, $Dur_{j,t}$ is an implicit function of $s_{j,t}$ and the VAR parameter estimates.

I estimate the VAR parameters on an expanding window such that $Dur_{j,t}$ only requires information that is publicly available by time t . The next section provides the empirical details on how I estimate the VAR parameters and construct portfolios sorted on equity duration.

2.2. Sample construction

This section details the sample construction.

2.2.1. Data

Stock return data come from the Center for Research in Security Prices (CRSP) monthly stock file and accounting data from Compustat annual file. I follow the literature

⁴ In the Internet Appendix, I provide the empirical density function of $dr_{j,t}$ obtained from all firm-year observations in my sample. The overall distribution is reasonable and concentrates between 5% and 20%. To keep the current paper focused on equity duration, I leave a more thoughtful exploration of $dr_{j,t}$ to future work.

and restrict my analysis to common stocks of firms incorporated in the United States ($shrcd = 10$ or 11) trading on NYSE, Amex, or Nasdaq ($exchcd = 1, 2$ or 3). I exclude utilities ($4900 \leq SIC \leq 4949$) and financials ($6000 \leq SIC \leq 6999$) and require a minimum of two previous years in Compustat for a company to be included in my analysis with the objective of alleviating backfilling concerns (see Fama and French, 1993).

To estimate equity duration, I form $s_{j,t}$ based on 12 state variables split among four broad categories:

(i) Valuation measures:

- book to market: $bm_{j,t} = \log(BE_{j,t}/ME_{j,t})$
- payout yield: $POy_{j,t} = \log(1 + PO_{j,t}/ME_{j,t})$
- sales yield: $Yy_{j,t} = \log(Y_{j,t}/ME_{j,t})$

(ii) Growth measures:

- book equity growth: $BEg_{j,t} = \log(BE_{j,t}/BE_{j,t-1})$
- asset growth: $Ag_{j,t} = \log(A_{j,t}/A_{j,t-1})$
- sales growth: $Yg_{j,t} = \log(Y_{j,t}/Y_{j,t-1})$

(iii) Profitability measures:⁵

- clean surplus profitability:
 $CSprof_{j,t} = \log\left(1 + \frac{CSE_{j,t}}{BE_{j,t-1}}\right)$
- return on equity: $Roe_{j,t} = \log\left(1 + \frac{E_{j,t}}{0.5BE_{j,t} + 0.5BE_{j,t-1}}\right)$
- gross profitability: $Gprof_{j,t} = \log\left(1 + \frac{GP_{j,t}}{0.5A_{j,t} + 0.5A_{j,t-1}}\right)$

(iv) Capital structure measures:

- market leverage: $Mlev_{j,t} = B_{j,t}/(ME_{j,t} + B_{j,t})$
- book leverage: $Blev_{j,t} = B_{j,t}/A_{j,t}$
- cash holdings: $Cash_{j,t} = C_{j,t}/A_{j,t}$

where ME is market equity from CRSP and all other variables are constructed from Compustat. BE is book equity following Davis, Fama, and French (2000); A is total assets (at); Y is total revenue ($revt$); PO is net payout following Boudoukh et al. (2007);⁶ E is income before extraordinary items (ib); GP is gross profits ($revt - cogs$) following

⁵ For the denominator of return on equity (gross profitability), I use the average of initial and final book equity (total assets) over the fiscal period. This is a compromise between the two typical approaches of using either beginning of period (e.g., Hou, Xue, and Zhang, 2015) or end of period (e.g., Novy-Marx, 2013; Fama and French, 2015) book equity or total assets to measure profitability. Profits are generated over the fiscal year so that neither the beginning nor the end of the period represents the basis for the profit generation, and thus I take the average between them. The definition of $CSprof$ in the calculation of duration does not allow me to use this approach for clean surplus profitability, which requires beginning of period book equity.

⁶ To define net payouts, PO , I start by defining book value of preferred stock ($BVPS$), which is given by redemption ($pstkrrv$), liquidation ($pstkli$), or par value ($pstk$) of preferred stock in this order. Then, the net payout in any given fiscal year is equal to cash dividends (dvc) + net equity repurchases, which is given by the total expenditure on the purchase of common and preferred stocks ($prstk$) - sale of common and preferred stock ($sstk$) + net issuances of preferred stocks ($BVPS_{j,t} - BVPS_{j,t-1}$). The Compustat data required to calculate net equity repurchases are only available starting in 1971. As such, for the earlier period, which is only used for the VAR estimation, I also follow Boudoukh et al. (2007) and use CRSP information on prices ($P_{j,t}$) and shares outstanding ($N_{j,t}$), $PO_{j,t} = P_{j,t} \cdot (N_{j,t} - N_{j,t-1})$.

Novy-Marx (2013); B is total book debt ($dltt + dlc$ as long as one of the two is available); and C is cash and short-term investments (che).⁷ All raw level quantities are deflated by the Consumer Price Index (CPI) before calculating ratios.

From Eq. (7), the state variables only matter for the duration calculation to the extent that they predict $CSprof$ or BEg at some horizon. Since these variables are the only source of cross-sectional variation in duration in my framework, 12 state variables is a reasonable compromise between parsimony and achieving cross-sectional variability in duration. In the Internet Appendix, I explore the short duration premium after dropping each state variable in $s_{j,t}$ one at a time and find results generally consistent with the ones reported in the main text.

2.2.2. Portfolios sorted on equity duration

At June of year t (with t from 1973 to 2017), I form ten (value-weighted and equal-weighted) decile portfolios by sorting stocks based on their $Dur_{j,t}$ and study the returns of these portfolios over the subsequent 12 months (so that portfolio returns go from July/1973 to June/2018). For value-weighted portfolios, I use NYSE breakpoints to define thresholds to assign stocks into portfolios but form portfolios with all stocks in the sample (as in Fama and French, 1993). For equal-weighted portfolios, I follow Hou, Xue, and Zhang (2019) and completely exclude microcaps (defined as the firms below the 20% quantile of market equity based on NYSE breakpoints) to make sure results are not due to these firms. I then define breakpoints and form decile portfolios with the remaining firms. For both value- and equal-weighted portfolios, I hold each stock for one year and repeat the procedure the following June (equal-weighted portfolios are rebalanced every month to keep equal weights for each stock). After stock delistings, I rebalance the portfolios to keep value- or equal-weighted returns across the available stocks.

$Dur_{j,t}$ in June of year t is estimated from Eq. (7) with $dr_{j,t}$ implied by the valuation identity (Eq. (6)) given the state vector, $s_{j,t}$. Accounting information used to construct the state vector is from the fiscal year ending in calendar year $t - 1$ and market equity as of December of $t - 1$ (same time convention as in Fama and French, 1992).

$Dur_{j,t}$ also depends on the VAR parameters, which are estimated using an expanding window such that estimates used in June of year t rely on information no later than December of calendar year $t - 1$. The first use of $Dur_{j,t}$ to create equity duration portfolios is in 1973, which gives ten years of data (1963–1972) with $s_{j,t}$ and $s_{j,t-1}$ available

⁷ I impose some minor screenings in the data used to get the state vector, $s_{j,t}$. I set any nonpositive A , BE , ME , and Y to missing as well as any negative C , B , and cash dividends. I also set to missing any BE , C , and B higher than A . Similar to Vuolteenaho (2002), I set to missing any BE higher than $50 \times ME$ or below $(1/50) \times ME$ and set any profitability ratio to -99% when below this value so that the log transformation is always feasible and firms do not lose more than 100% their book equity (or assets). Finally, to avoid the effect of outliers, I winsorize each non-bounded variable in the state vector at 1% and 99% percentiles for each cross-section (this avoids any look-ahead bias in the winsorization).

for the initial estimation of the VAR parameters as I do not rely on Compustat data before 1962.⁸

I use the same Γ and Σ for all firms so that all cross-sectional variability in duration comes from the state vector, $s_{j,t}$. I estimate the autoregressive matrix, Γ , equation by equation from Fama-MacBeth regressions and the covariance matrix, Σ , based on the sample analogue constructed from pooling observations of firm-demeaned residuals.⁹ To minimize the effect of small stocks on the VAR parameters, I always exclude microcaps (firms below the 20% quantile of market equity based on NYSE breakpoints) when estimating Γ and Σ even when microcaps are included in the portfolios. The intercepts in the Γ matrix define the long-term behavior of $\mathbb{E}[CSProf]$ and $\mathbb{E}[BEg]$ and have no cross-sectional variability given the use of a common Γ across firms. To minimize the effect of extreme observations on the long-term expected profitability and growth, I select the intercepts so that the long-term expectation of $s_{j,t}$ matches the time series average of cross-sectional medians for each of the variables in the state vector (using the same expanding window as for other parameters in Γ).

3. Portfolios sorted on equity duration

In this section, I study the basic properties of the short duration premium. Section 3.1 reports summary statistics on the sample I use to construct the duration portfolios. Section 3.2 validates the equity duration measure I rely on. Section 3.3 demonstrates the short duration premium is (economically and statistically) significant, long-lived, and strong even among the largest firms. Section 3.4 shows that the value and profitability premia disappear after controlling for duration.

3.1. Summary statistics for stocks in duration portfolios

Table 1 summarizes the sample of firms used to construct the duration portfolios (Panel A) as well as the characteristics of firms in each portfolio (Panel B).¹⁰

The first two columns of Panel A demonstrate that the sample is comprehensive despite requiring the availability of the full state vector, $s_{j,t}$. Specifically, the average number of firms available in a given year is 2,554, which, on average, accounts for 88.2% of the market equity available in an analogous sample that requires only *ME* and *BE* to be nonmissing. The other columns show the year-by-year

cross-sectional distribution of some of the key variables in my analysis: duration (*Dur*), book to market (*BM*), book equity growth (*BEg*), and clean surplus profitability (*CSprof*).

There is substantial variability in duration, which is important since my goal is to form portfolios that are as distinct as possible on this dimension. The time series average of the median equity duration is 38.9 years, with the 10% and 90% quantiles being 17.1 and 94.7 years, respectively. This equity duration distribution is reasonable as it implies plausible variation in average dividend yields in the context of a simple log-linear approximation. Specifically, Campbell and Shiller (1989)'s log-linear approximation implies $Dur \approx 1 + e^{-\bar{d}p}$, where $\bar{d}p$ is the firm log dividend yield.¹¹ The average $\bar{d}p$ obtained by aggregating firm-level information over my sample yields $Dur \approx 48.6$ years. The same exercise implies $Dur \approx 38.1$ years if we replace dividends with net payout. Moreover, a stock with a relatively high average dividend yield (of 6%) would have a low equity duration of 17.67 years, and a stock with a relatively low dividend yield (of 1%) would have a high equity duration of 101 years.

Panel B displays the time series averages of portfolio-level characteristics (value-weighted across firms). The short duration decile tends to be composed of small, value, low growth, and profitable companies. The opposite is true for the long duration decile (albeit not all relations are monotone). I further explore the relation between duration, size, valuation, growth, and profitability in Section 3.4.

3.2. Validating duration portfolios

Fig. 1 validates the duration portfolios I construct by demonstrating that (i) shorter duration stocks pay a larger fraction of their market equity over the short term (one to ten years) and (ii) longer duration stocks are more exposed to movements in the dividend term structure as defined by the return differential between the equity index and its short-term dividend claim (for a discussion, see Binsbergen and Kojien, 2017).

Fig. 1 (a) shows the cumulative fraction of market equity that firms in each duration decile pay (in net payouts) over the ten years following the duration measurement. Fig. 1(b) repeats this analysis after replacing net payouts with cash dividends.¹² Focusing on net payouts, firms in the short duration portfolio tend to pay, on average, about

⁸ One might worry that the VAR parameter estimates are not stable since the first estimate uses only ten years of data. In the Internet Appendix, I show that this is not the case as the out-of-sample VAR parameter estimates tend to be fairly close to the VAR parameter estimates obtained using the entire sample period. The intuition is that the VAR estimation relies on the entire cross-section of stocks so that even a short time period (such as ten years) provides a large number of observations for the estimation.

⁹ As it is common in the literature, I weight each cross-section in the Fama-MacBeth regression by the number of firms in that cross-section. Also, I construct s_t using market equity as of the end of each accounting fiscal year when estimating the VAR to keep a one-year interval between $s_{j,t}$ and $s_{j,t-1}$ during the estimation.

¹⁰ For completeness, rank correlations between firm-level characteristics are also provided in the Internet Appendix.

¹¹ Replacing all expected returns by the same discount rate, dr_t , in Campbell and Shiller (1989)'s log-linear approximation to stock prices yields: $p_t = \text{constant} + d_t + \sum_{h=1}^{h=\infty} \rho^{h-1} (\mathbb{E}_t[\Delta d_{t+h}] - dr_t)$. Then, duration is given by $-\partial p_t / \partial dr_t = \sum_{h=1}^{h=\infty} \rho^{h-1} = 1/(1-\rho) = 1 + e^{-\bar{d}p}$, with the last equality using $\rho = 1/(1 + e^{\bar{d}p})$.

¹² Specifically, for portfolios formed in June of year $t+1$, I start by restricting the sample to firms with fiscal year end in December of year t to align all accounting information. Then, I measure the total market equity as of December of year t for each portfolio as well as the total net payout (and cash dividends) paid in years $t+1, t+2, \dots, t+10$ (with fixed portfolio composition). The fraction of *ME* paid in year $t+\tau$ is $\sum_{j=1}^{N_{t+\tau}} PO_{j,t+\tau} / \sum_{j=1}^{N_{t+\tau}} ME_{j,t}$, and I sum these values for $1 \leq \tau \leq h$ to obtain the fraction of *ME* paid within h years (with analogous procedure for cash dividends). The graphs report the time series average of these fractions for each portfolio for $h = 1, 2, \dots, 10$.

Table 1

Summary statistics for firms in duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on *Dur*, which is measured from Eq. (7) (empirical details in Section 2). Panel A reports sample statistics at June of selected years for all firms included in the duration portfolios. *N* is the total number of sample firms; *ME* represents the percentage of market equity in my sample relative to a comparable sample that requires only *ME* and *BE* availability; and q_{pk}^x represents the *p*-th quantile of variable *x* based on the respective cross-section of firms. Panel B reports, for each portfolio, *Dur*, *Size* = $\log(\text{ME})$, and the firm characteristics in the VAR state vector (defined in Section 2.2). At each year, I take the value-weighted average of the respective characteristic within each duration portfolio and, at the end, take the time average of these aggregate characteristics separately for each portfolio. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994) with t_{stat} in parentheses.

Panel A: Sample of firms included in duration portfolios															
Year	N	%ME	q _{Dur10%}	q _{Dur50%}	q _{Dur90%}	q _{BM10%}	q _{BM50%}	q _{BM90%}	q _{BEG10%}	q _{BEG50%}	q _{BEG90%}	q _{CSprof10%}	q _{CSprof50%}	q _{CSprof90%}	
1973	1,507	91.0%	5.2	19.5	55.3	0.3	0.80	1.70	−0.05	0.053	0.216	−0.042	0.077	0.190	
1978	2,540	97.0%	5.6	15.6	37.2	0.5	1.16	2.43	−0.10	0.038	0.183	−0.068	0.071	0.188	
1983	2,455	86.6%	10.3	29.7	66.7	0.3	0.85	1.91	−0.23	0.018	0.204	−0.232	0.049	0.188	
1988	2,690	87.5%	14.5	36.5	88.2	0.3	0.82	1.80	−0.33	0.037	0.346	−0.359	0.052	0.237	
1993	2,860	91.3%	19.3	46.4	103.8	0.2	0.59	1.54	−0.31	0.033	0.446	−0.408	0.043	0.267	
1998	3,308	82.5%	26.9	56.5	135.4	0.2	0.46	1.20	−0.32	0.071	0.418	−0.411	0.092	0.316	
2003	2,831	84.6%	15.3	42.7	100.7	0.2	0.73	2.05	−0.54	0.006	0.264	−0.604	0.019	0.242	
2008	2,402	87.4%	25.0	50.2	111.6	0.2	0.49	1.23	−0.33	0.048	0.332	−0.335	0.093	0.314	
2013	2,126	91.1%	19.7	42.5	102.6	0.2	0.56	1.40	−0.31	0.032	0.249	−0.337	0.084	0.274	
2017	1,922	89.3%	25.1	50.8	136.3	0.1	0.42	1.12	−0.38	0.000	0.222	−0.431	0.058	0.279	
Average	2,554	88.2%	17.1	38.9	94.7	0.2	0.73	1.79	−0.28	0.034	0.309	−0.325	0.063	0.259	
Panel B: Characteristics of firms in each duration portfolio															
Duration				Valuation			Growth			Profitability			Capital Structure		
decile	Dur	Size	BE/ME	PO/ME	Y/ME	BEg	Ag	Yg	CSprof	Roe	Gprof	Mlev	Blev	Cash	
Short	16.5	7.7	1.42	0.045	3.98	0.016	0.003	0.012	0.072	0.100	0.381	0.287	0.171	0.114	
2	24.7	8.4	0.95	0.041	2.53	0.037	0.027	0.031	0.105	0.127	0.383	0.238	0.183	0.112	
3	29.9	8.8	0.77	0.037	1.89	0.050	0.047	0.046	0.129	0.142	0.380	0.213	0.187	0.116	
4	34.4	9.3	0.66	0.040	1.58	0.047	0.043	0.043	0.138	0.153	0.368	0.197	0.193	0.117	
5	38.5	9.7	0.58	0.038	1.32	0.054	0.051	0.052	0.143	0.154	0.346	0.183	0.195	0.118	
6	42.9	9.9	0.50	0.035	1.06	0.060	0.065	0.057	0.156	0.164	0.353	0.178	0.206	0.115	
7	48.1	10.0	0.44	0.030	0.91	0.060	0.067	0.066	0.154	0.162	0.338	0.179	0.218	0.113	
8	54.8	9.9	0.41	0.024	0.77	0.064	0.078	0.074	0.139	0.142	0.312	0.186	0.223	0.112	
9	66.9	9.5	0.35	0.014	0.71	0.105	0.131	0.105	0.152	0.125	0.284	0.210	0.259	0.106	
Long	105.0	9.4	0.29	0.003	0.75	0.100	0.171	0.126	0.061	−0.031	0.223	0.285	0.308	0.117	
L-S	88.4	1.7	−1.12	−0.043	−3.24	0.084	0.167	0.114	−0.012	−0.131	−0.157	−0.002	0.137	0.003	
(t _{L-S})	(14.38)	(6.06)	(−6.19)	(−8.53)	(−5.76)	(2.76)	(8.03)	(7.58)	(−0.35)	(−3.66)	(−3.64)	(−0.04)	(5.67)	(0.32)	

35% of their market equity over the first ten years after duration is measured. The same figure is close to 15% for firms in the long duration portfolio. The evidence indicates that firms classified as short duration equity indeed have a larger fraction of equity value associated with short-term cash flows in comparison to firms classified as long duration equity.

Fig. 1 (c) and (d) show the regression slopes of duration portfolio returns on the dividend term structure, measured as returns on a long-short portfolio that buys a long duration asset (an equity index) and sells a short duration asset (a short-term dividend claim). Fig. 1(c) uses returns on the S&P 500 as the long-duration asset and returns on a S&P 500 dividend claim (with maturity between one and two years) as the short duration asset.¹³ Fig. 1(d) provides results using (shocks to) annual log returns on the CRSP equity portfolio as the long duration asset and (shocks to)

annual log dividend growth on the same portfolio as the short duration asset.¹⁴ This alternative approach captures the dividend term structure because (shocks to) annual log returns on a one-year dividend claim, $R_{d,t}^{(1)} = D_t/P_{t-1}^{(1)}$, are equal to (shocks to) annual log dividend growth: $\log(R_{d,t}^{(1)}) - \mathbb{E}_{t-1}[\log(R_{d,t}^{(1)})] = \Delta d_t - \mathbb{E}_{t-1}[\Delta d_t]$.

Consistent with the idea that cash flow duration increases from the first to the tenth duration decile, the exposure of duration portfolios to the dividend term structure tends to increase as we move from decile one to decile ten. This is true regardless of whether we focus on value- or equal-weighted portfolios.

The overall evidence indicates that my equity duration measure indeed captures equity cash flow duration. This result does not imply my duration measure is perfect. For instance, the relative cash flow difference between short

¹³ To avoid potential illiquidity/microstructure issues, I follow the recommendation in Boguth et al. (2012) and use annual log returns for the long-short strategy (and rely on annual log excess returns on the duration portfolios to measure betas). Annual returns on the S&P 500 dividend claim go from June of 1997 to June of 2018 (see the Internet Appendix for a detailed description of its construction).

¹⁴ Shocks are measured as residuals on annual predictive regressions in which the predictive variables are the ones in the aggregate state vector described in the next section, but results are not sensitive to changing this specification. For instance, using only dividend yield as a predictive variable delivers very similar results. Further details on the measurement of returns, dividend growth, and the aggregate state variables are provided in Section 4.

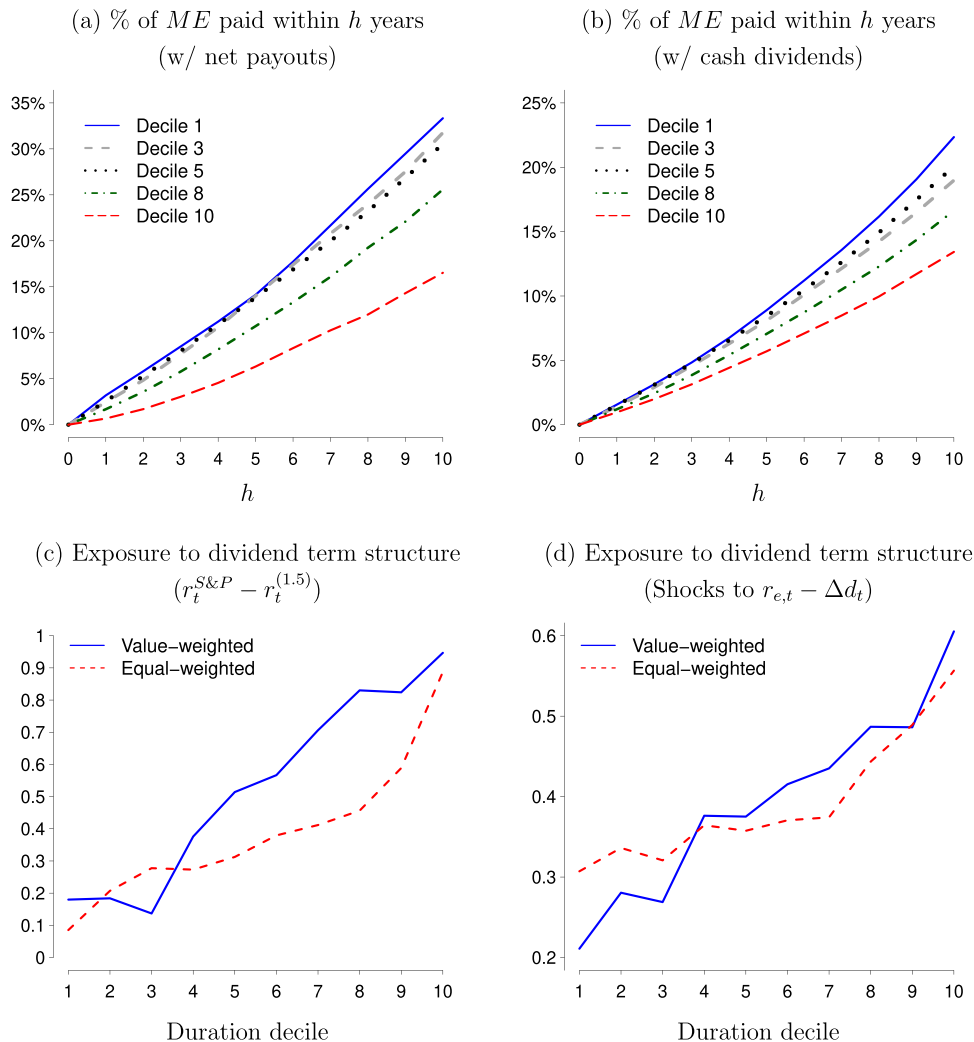


Fig. 1. Validating equity duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from Eq. (7) (empirical details in Section 2). Graph (a) shows the cumulative fraction of market equity that firms in each duration decile pay (in net payouts) over the ten years following the duration measurement. Graph (b) repeats this analysis after replacing net payouts with cash dividends. Graphs (c) and (d) show the exposures (regression slopes) of duration portfolios (value-weighted and equal-weighted) to the dividend term structure, measured as returns on a long-short portfolio that buys a long duration asset (an equity index) and sells a short duration asset (a short-term dividend claim). Graph (c) uses annual log returns on the S&P 500 as the long duration asset and annual log returns on a S&P 500 dividend claim (with approximately 1.5 year maturity) as the short duration asset (first annual return is in June/1997). Graph (d) provides results using (shocks to) annual log returns on the CRSP equity portfolio as the long duration asset and (shocks to) annual log dividend growth on the same portfolio as the short duration asset. All necessary details are provided in Section 3.2.

and long duration firms may not induce a true duration difference as large as 88.4 years between the short and long duration portfolios as suggested by the Dur measure (see Table 1, Panel B). However, for the purpose of this paper, only the duration rank matters (as opposed to the specific duration values), and the evidence indicates Dur provides a reasonable equity duration ranking of firms.

3.3. The short duration premium

Table 2 shows the existence of a short duration premium; a strategy that buys the short duration decile and sells the long duration decile has positive average returns

from July of 1973 to June of 2018. Moreover, the premium (i) lasts for at least five years, (ii) is large even among the largest US firms, (iii) delivers a substantial Sharpe ratio, and (iv) remains strong (as α) after accounting for exposure to standard risk factors in the literature.

Panel A focuses on (annualized) average excess returns and Sharpe ratios. I focus the discussion on value-weighted returns (first half of the panel), but results are slightly stronger for equal-weighted returns (second half of the panel). Column $\bar{r}_{t \rightarrow t+1}$ shows a short duration premium of 8.6% on an annual basis ($t_{stat} = 3.55$). Moreover, the decline in average excess returns from decile one to decile ten is almost monotonic, and a similar pattern is observed in

Table 2

Performance of duration portfolios: the short duration premium.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on *Dur*, which is measured from Eq. (7) (empirical details in Section 2), and portfolio returns are measured monthly. Panel A shows annualized average excess returns ($\times 12$) and Sharpe ratios ($\times \sqrt{12}$). $\bar{r}_{t+j \rightarrow t+h}$ indicates the portfolio is implemented j years after the *Dur* measurement and is held for $h - j$ years. Moreover, $\bar{r}_{t \rightarrow t+1}^{large}$ indicates the portfolio is based on firms with market equity above the 80% NYSE quantile, and $\bar{r}_{t \rightarrow t+1}^{dlst adj}$ indicates returns are adjusted for delistings. Panel B reports β s and annualized α s ($\times 12$) from factor regressions. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994) with t_{stat} in parentheses and p -values in brackets.

Panel A: Average returns and Sharpe ratios													
Duration	Value-weighted portfolios						Duration	Equal-weighted portfolios					
	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{large}$	$\bar{r}_{t \rightarrow t+1}^{dlst adj}$	$\bar{r}/\sigma$		$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{large}$	$\bar{r}_{t \rightarrow t+1}^{dlst adj}$	$\bar{r}/\sigma$
Short	12.7%	11.8%	11.8%	10.3%	12.7%	0.66	Short	12.9%	12.7%	12.0%	11.0%	12.9%	0.64
2	11.5%	11.0%	10.5%	10.1%	11.5%	0.66	2	14.2%	13.2%	11.8%	9.8%	14.3%	0.73
3	12.5%	11.6%	11.3%	9.9%	12.6%	0.72	3	12.9%	12.9%	12.8%	11.3%	12.8%	0.68
4	11.6%	10.2%	8.1%	11.8%	11.6%	0.71	4	12.4%	12.2%	11.7%	11.7%	12.4%	0.65
5	11.0%	9.2%	7.5%	5.5%	11.0%	0.67	5	11.2%	12.0%	12.0%	7.7%	11.2%	0.59
6	8.8%	8.7%	8.5%	6.7%	8.8%	0.55	6	11.2%	11.2%	11.1%	8.4%	11.2%	0.57
7	8.1%	7.5%	7.2%	6.0%	8.1%	0.50	7	9.7%	10.6%	11.0%	7.3%	9.7%	0.49
8	7.3%	7.0%	7.5%	6.9%	7.3%	0.42	8	8.8%	9.6%	10.9%	6.9%	8.8%	0.43
9	5.5%	6.5%	7.5%	3.8%	5.6%	0.30	9	6.5%	7.8%	8.6%	4.9%	6.5%	0.29
Long	4.1%	4.7%	7.7%	3.7%	4.2%	0.20	Long	3.6%	6.1%	8.1%	5.3%	3.3%	0.13
L-S	−8.6%	−7.1%	−4.1%	−6.6%	−8.6%	−0.57	L-S	−9.4%	−6.6%	−3.9%	−5.7%	−9.6%	−0.61
(t_{L-S})	(−3.55)	(−3.82)	(−2.03)	(−2.31)	(−3.53)	[0.00]	(t_{L-S})	(−3.32)	(−2.76)	(−1.56)	(−2.18)	(−3.34)	[0.00]
Panel B: Risk-adjusted performance based on factor models													
Duration	CAPM		Fama and French (2015) 5-factors						Hou, Xue, and Zhang (2015) q-factors				
	α_{CAPM}	β_{MKT}	α_{FF}	β_{MKT}	β_{SMB}	β_{HML}	β_{CMA}	β_{RMW}	α_q	β_{MKT}	β_{SIZE}	β_{INV}	β_{ROE}
Value-weighted portfolios													
Short	5.1%	0.97	0.4%	0.99	0.67	0.33	0.14	0.21	2.7%	0.94	0.55	0.42	−0.15
2	4.6%	0.94	1.3%	0.95	0.46	0.19	0.09	0.19	2.6%	0.92	0.39	0.22	−0.01
3	5.4%	0.97	1.8%	1.01	0.37	0.17	0.11	0.32	2.9%	0.97	0.29	0.24	0.08
4	4.8%	0.93	2.6%	0.97	0.17	0.14	0.08	0.15	3.7%	0.94	0.13	0.16	0.02
5	4.3%	0.95	2.5%	0.98	0.12	−0.09	0.30	0.14	2.8%	0.96	0.11	0.10	0.11
6	2.2%	0.91	0.6%	0.95	0.08	−0.10	0.22	0.19	0.9%	0.93	0.06	0.07	0.13
7	1.1%	0.95	0.3%	0.97	0.00	−0.08	0.08	0.19	0.7%	0.97	−0.04	0.01	0.10
8	−0.2%	1.02	0.2%	1.02	−0.06	−0.16	0.13	−0.02	0.6%	1.02	−0.08	−0.04	−0.01
9	−2.6%	1.10	−3.0%	1.11	0.03	−0.10	0.04	0.14	−1.8%	1.10	−0.05	−0.09	0.01
Long	−4.9%	1.25	−4.1%	1.20	0.13	−0.14	−0.05	−0.04	−3.7%	1.21	0.10	−0.19	−0.02
L-S	−10.0%	0.28	−4.4%	0.21	−0.55	−0.47	−0.19	−0.25	−6.4%	0.27	−0.45	−0.61	0.13
(t_{L-S})	(−3.78)	(4.20)	(−2.49)	(2.98)	(−4.62)	(−2.77)	(−1.20)	(−2.39)	(−2.86)	(4.24)	(−3.61)	(−3.63)	(0.89)
Equal-weighted portfolios													
Short	4.8%	1.06	−1.2%	1.07	0.90	0.42	0.10	0.33	0.8%	1.01	0.77	0.55	−0.10
2	6.3%	1.07	2.1%	1.05	0.79	0.32	0.00	0.22	4.0%	1.01	0.67	0.30	−0.12
3	4.7%	1.09	1.5%	1.05	0.67	0.22	0.02	0.14	3.0%	1.02	0.58	0.19	−0.09
4	4.4%	1.09	1.2%	1.07	0.62	0.16	0.05	0.21	2.6%	1.03	0.53	0.16	−0.03
5	3.1%	1.10	0.2%	1.07	0.61	0.15	0.03	0.15	1.9%	1.03	0.52	0.13	−0.10
6	2.8%	1.14	0.7%	1.09	0.58	0.08	−0.03	0.15	2.1%	1.06	0.49	0.02	−0.08
7	1.2%	1.16	0.0%	1.10	0.50	0.09	−0.10	0.05	1.5%	1.08	0.42	−0.06	−0.12
8	−0.1%	1.21	−1.9%	1.16	0.50	−0.04	0.10	0.11	−0.5%	1.14	0.41	0.01	−0.07
9	−3.1%	1.30	−4.0%	1.21	0.57	−0.10	0.03	0.02	−2.6%	1.21	0.47	−0.10	−0.14
Long	−7.5%	1.51	−5.8%	1.33	0.60	−0.09	−0.20	−0.34	−3.2%	1.35	0.47	−0.30	−0.50
L-S	−12.3%	0.45	−4.6%	0.26	−0.30	−0.51	−0.31	−0.68	−4.0%	0.33	−0.30	−0.84	−0.39
(t_{L-S})	(−4.12)	(5.31)	(−2.23)	(5.15)	(−3.18)	(−3.46)	(−2.00)	(−4.16)	(−1.81)	(6.37)	(−2.81)	(−4.74)	(−2.00)

Sharpe ratios (Column $\bar{r}/\sigma$). Column $\bar{r}_{t \rightarrow t+5}$ shows the general declining pattern in average returns from short to long duration stocks is also true if the portfolio holding period is five years, with an annual premium of 7.1% ($t_{stat} = 3.82$). Based on column $\bar{r}_{t+4 \rightarrow t+5}$, even if we use duration estimated four years before the sorting date, we still observe a short duration premium of 4.1% ($t_{stat} = 2.03$). Column $\bar{r}_{t \rightarrow t+1}^{large}$ shows the premium is still significant (6.6% with $t_{stat} = 2.31$) when only large firms are used to construct

portfolios (defined as the firms above the 80% quantile of market equity based on NYSE breakpoints). Finally, column $\bar{r}_{t \rightarrow t+1}^{dlst adj}$ shows the premium is basically identical after accounting for delistings (8.6% with $t_{stat} = 3.53$).¹⁵

¹⁵ To account for delistings, I use CRSP delisting data to calculate delisting returns whenever possible, and when a delisting return cannot be calculated, I assume a −30% return for delisting by cause ($400 \leq dlstd \leq 599$) based on the findings of Shumway (1997) and a 0% return for

Panel B presents the results of factor regressions. The first two columns show that the short duration premium is even larger when stated in terms of annualized CAPM α s (10.0% with $t_{stat} = 3.78$) since market betas increase in equity duration. The next set of columns show that the extra risk factors in the Fama and French (2015)'s five-factor model partially explain this premium, but the β patterns are relatively weak, nonmonotonic, and still leave an α of 4.4% ($t_{stat} = 2.49$), more than half the raw short duration premium. The last set of columns show relatively similar results using Hou, Xue, and Zhang (2015)'s q-factor model. The α is still 6.4% ($t_{stat} = 2.86$) with betas on the size and investment factors helping to reduce the short duration premium slightly.

Fig. 2 reports the short duration premium (value- and equal-weighted) on a 20-year rolling window to demonstrate that it is not concentrated in any particular period. Clearly, the short duration premium is positive for any 20-year window one selects. The figure also plots the 95% confidence interval. There are periods in which we cannot reject that the premium is zero, but this is a consequence of the higher standard errors associated with a shorter sample. Even in these periods, the estimated premium is way above 4%, which is substantial if we consider that these are the 20-year periods with the weakest short duration premia in the sample.

The relatively stable short duration premium does not imply the short duration strategy always delivers positive returns. Fig. 3 reports annual realized returns on the short duration strategy (decile one minus decile ten) and shows that it sometimes delivers largely negative returns. For instance, the short duration strategy performed poorly in the late 1990s during the tech boom and extremely well when the tech boom burst in the early 2000s.

3.4. Equity duration, value, and profitability

I now explore the interaction between duration and the four firm-level characteristics that have been included in recent factor models (e.g., Fama and French, 2015; Hou, Xue, and Zhang, 2015) as important determinants of expected returns: value, profitability, asset growth, and size.¹⁶ The key result is that the value and profitability premia disappear after controlling for duration.

Simultaneously controlling for multiple firm-level characteristics in asset pricing tests is challenging. As Cochrane (2011) states, “we cannot chop portfolios 27

ways, so I think we will end up running multivariate regressions...Running multiple panel-data forecasting regressions is full of pitfalls of course. One can end up focusing on tiny firms, or outliers. One can get the functional form wrong...we must address the factor zoo, and I do not see how to do it by a high-dimensional portfolio sort.”

Since my goal is to study the effect of duration, value, and profitability jointly, I design a simple method to compare multiple characteristics in the context of panel regressions while still relying on portfolio sorts to address Cochrane (2011)'s concerns with stock-level panel regressions. Specifically, I estimate panel regressions using portfolio returns on the left side and the average decile values as covariates.

Consider the case in which only duration and value are studied. I form 10 decile portfolios for duration and 10 for value and assign each stock a duration decile number as well as the value decile number (i.e., I create value decile and duration decile as firm-level characteristics). Then, for each duration portfolio, I calculate the weighted average value decile of its stocks and similarly for value portfolios. At the end of this procedure, I have 20 decile portfolios with each portfolio having an average duration decile as well as an average value decile. I then regress portfolio returns on portfolio deciles, which is a multivariate version of the typical average high-low returns.¹⁷

Table 3 provides results from these panel regressions of returns on portfolio deciles. The description focuses on value-weighted portfolios, but equal-weighted results are similar. Columns 1.1 to 1.8 only include in the panel data the decile portfolios of characteristics accounted for in the respective regression specification. For instance, column 1.2 uses 20 deciles, 10 from duration and 10 from book to market. Columns 2.1 to 2.7 keep all 50 decile portfolios (for duration, value, asset growth, profitability, and size) irrespective of the regression specification.

Column 1.1 shows univariate regression results. The implied short duration premium is 8.6% ($t_{stat} = 3.85$) on an annual basis, which is very similar to the findings in Table 2. All other variables induce premia with the expected sign. Only the profitability premium is insignificant in value-weighted portfolios ($t_{stat} = 0.70$), but it is still significant when considering equal-weighted portfolios ($t_{stat} = 2.12$).

Columns 1.2 to 1.5 consider the premium associated with each variable after controlling for duration. For all

other cases. I pro-rate delisting returns over the period from the last available price to the month of the delisting price as in Cohen, Polk, and Vuolteenaho (2009).

¹⁶ Following Fama and French (1992), I use book to market to proxy for value and market equity to proxy for size. My asset growth measurement is consistent with Fama and French (2015) and Hou, Xue, and Zhang (2015), and I proxy for profitability using the gross profitability measure in Novy-Marx (2013) as he finds that gross profitability performs better than many other profitability variables in predicting the cross-section of stock returns. To be consistent with Novy-Marx (2013), I use current assets as the denominator of gross profitability as opposed to $0.5 \cdot A_{t-1} + 0.5 \cdot A_t$ (used in the VAR to account for the fact that profits are generated over the fiscal year period). I still refer to the gross profitability measure as *Gprof*, and the results are very similar if the measurement is set to be consistent with the VAR estimation.

¹⁷ Specifically, with only one covariate (e.g., only ten duration deciles), the OLS estimate, $\hat{b}$, of this pooled panel regression is given by $9 \cdot \hat{b} = \sum_{i=0}^4 w_i \cdot (\bar{R}_{10-i} - \bar{R}_{1+i}) \cdot \frac{9}{9-2i}$, where $w_i = (4.5 - i)^2 / \sum_{k=0}^4 (4.5 - k)^2$ are weights and $9 \cdot \hat{b}$ provides an estimate for the high-low portfolio predicted by the given regression model since there are nine decile increases between deciles one and ten. The predicted premium is a weighted average of several terms. The first term is the average return on the high-low portfolio (when $i = 0$, we have $\bar{R}_{10} - \bar{R}_1$), which is the focus of the typical analysis. The second term, $(9/7) \cdot (\bar{R}_9 - \bar{R}_2)$, is the spread between the second set of most extreme portfolios scaled to have the same units as average return on the high-low portfolio. All other terms are similar, with the last term being $9 \cdot (\bar{R}_6 - \bar{R}_5)$. Of course, not all long-short portfolios provide the same level of information, with the most extreme deciles being more important. The OLS weights the terms accordingly (through w_i) to incorporate information from all decile spreads in the appropriate manner.

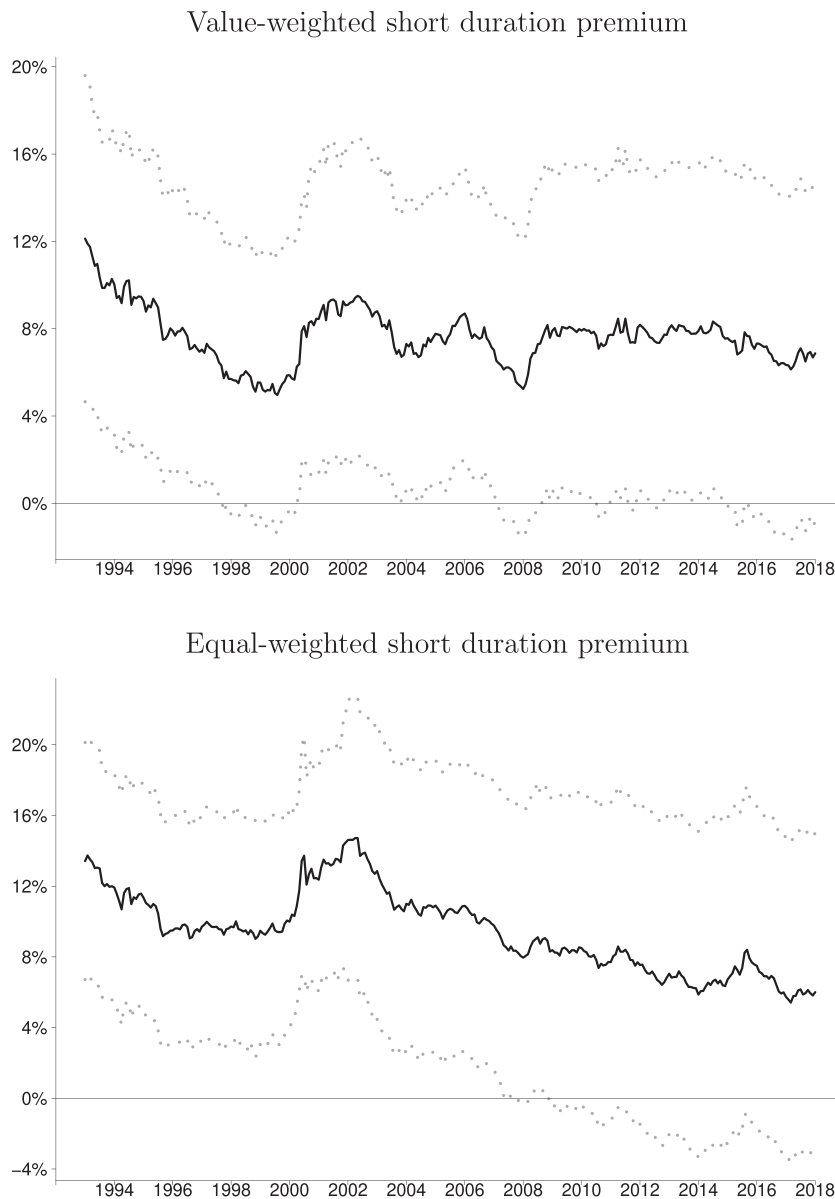


Fig. 2. Short duration premium on a 20-year rolling window.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from Eq. (7) (empirical details in Section 2). The graphs report the short duration premium (decile 1 minus decile 10) on a rolling window of 20 years (solid line). Year t on the horizontal axis is aligned with June of year t . The 95% confidence interval for the duration premium is also reported (dotted lines).

variables other than duration, the premium is reduced and becomes insignificant, with the exception that the premium for low asset growth companies is reduced but still significant in equal-weighted portfolios ($t_{stat} = 2.62$). The value premium is substantially reduced after accounting for duration, changing from a significant 4.9% ($t_{stat} = 2.06$) to an insignificant 0.9% ($t_{stat} = 0.32$). Moreover, the point estimate for the profitability premium even becomes negative. In contrast to the unrobust value, profitability, asset growth, and size premia, the short duration premium tends to become stronger after controlling for other premia. For instance, after controlling for the profitability premium, the short duration premium becomes 12.3% ($t_{stat} = 4.28$).

One of the key messages in Novy-Marx (2013) is that the value and profitability premia become substantially stronger after controlling for each other. Column 1.6 demonstrates this result. The value premium increases from 4.9% ($t_{stat} = 2.06$) to 12.7% ($t_{stat} = 3.33$), and the profitability premium increases from an insignificant 1.5% ($t_{stat} = 0.70$) to a significant 10.0% ($t_{stat} = 2.84$).

Interestingly, after controlling for duration, the value and profitability premia disappear despite also being controlled for each other. In fact, the point estimates for both premia even become negative in value-weighted portfolios. Specifically, the value premium decreases from

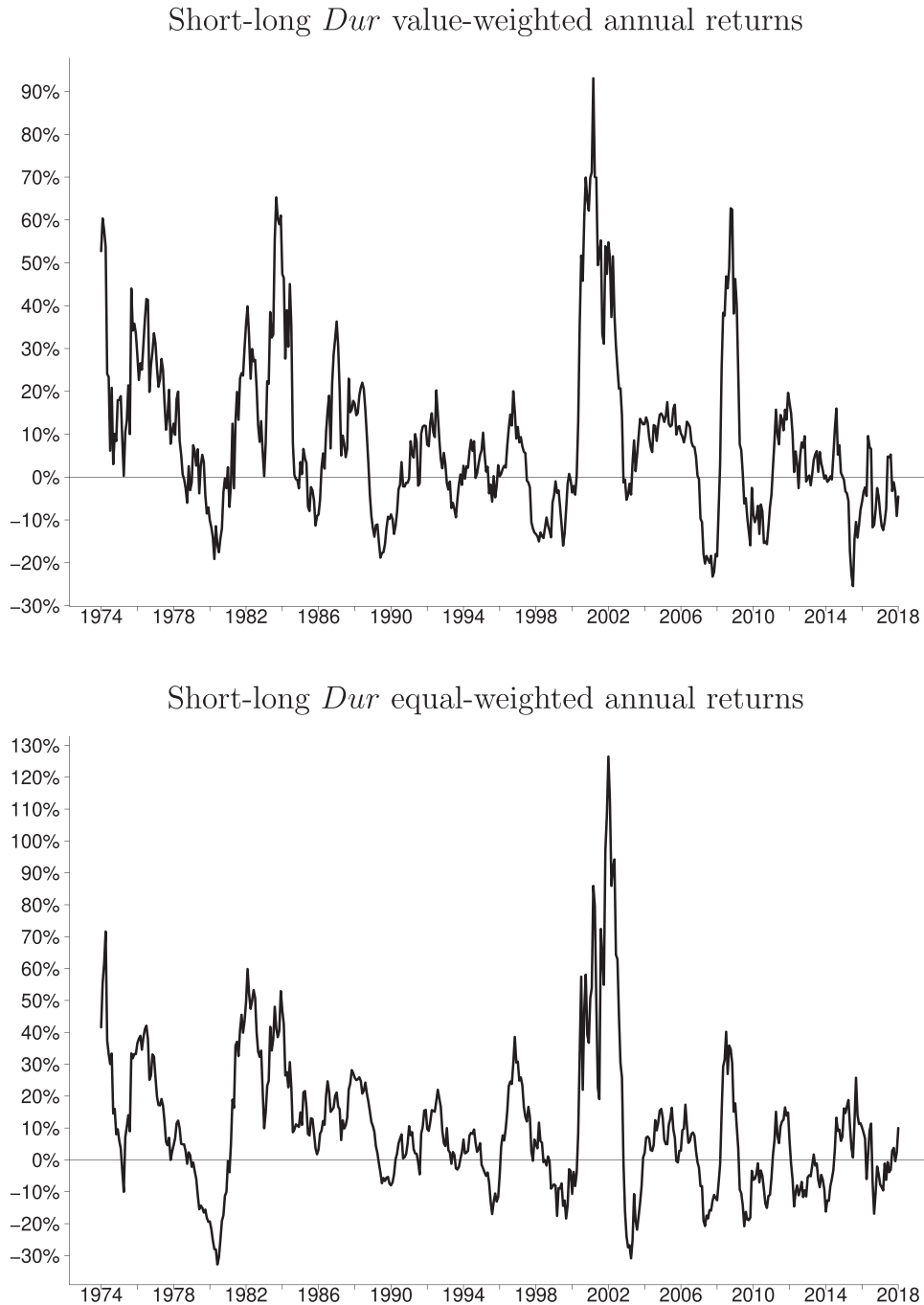


Fig. 3. Short duration premium on a one-year rolling window.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on *Dur*, which is measured from Eq. (7) (empirical details in Section 2). The graphs report monthly observations of annual realized returns on the short duration strategy (decile 1 minus decile 10). Year t on the horizontal axis is aligned with June of year t .

12.7% ($t_{stat} = 3.33$) to -0.4% ($t_{stat} = -0.09$), and the profitability premium decreases from 10.0% ($t_{stat} = 2.84$) to -1.6% ($t_{stat} = -0.34$). In contrast to the value and profitability premia, the short duration premium becomes very strong after controlling for both value and profitability (12.7% with $t_{stat} = 2.87$).

Finally, in a panel regression including all five variables, only the short duration premium is significant (14.4% with $t_{stat} = 2.44$) with the point estimates for the value and profitability premia remaining negative. Equal-weighted portfolios deliver similar qualitative results, except that the point estimates for the value and profitability premia

Table 3

Panel regressions of portfolio returns on portfolio deciles.

Portfolios are formed every June (1973 to 2017) from deciles based on the respective characteristics, which are described in Section 2. Monthly portfolio returns span the subsequent 12 months (from July/1973 to June/2018). The table reports results from panel regressions of portfolio returns on the lagged deciles for the respective variables, with coefficient units being comparable to annualized average returns of a high-low strategy (Section 3.4 provides details on the methodology). Columns 1.1 and 2.1 are based on univariate regressions, while all other columns rely on multivariate regressions using deciles from the firm-level characteristics. Each of the Columns 1.1 to 1.8 only includes in the panel data the decile portfolios of characteristics accounted for in the respective regression specification. For instance, column 1.2 uses 20 deciles, 10 from duration, and 10 from book to market. Columns 2.1 to 2.7 keep all fifty decile portfolios irrespective of the regression specification. t_{stat} are in parentheses, and statistical inference is based on the method in Driscoll and Kraay (1998), which is the natural generalization of Newey and West (1987, 1994) to a panel data setting and is robust to heteroskedasticity, autocorrelation, and cross-sectional correlation between portfolio returns.

Panel A: Value-weighted portfolios															
Sorting	Decile portfolios based on included covariates								All 50 decile portfolios						
variable	[1.1]	[1.2]	[1.3]	[1.4]	[1.5]	[1.6]	[1.7]	[1.8]	[2.1]	[2.2]	[2.3]	[2.4]	[2.5]	[2.6]	[2.7]
<i>Dur</i>	−8.6% (−3.85)	−9.7% (−4.22)	−12.3% (−4.28)	−10.1% (−3.98)	−9.7% (−4.26)		−12.7% (−2.87)	−14.4% (−2.44)	−14.5% (−4.83)	−13.8% (−4.24)	−14.6% (−4.65)	−13.6% (−4.29)	−13.6% (−4.40)		−14.2% (−2.45)
<i>BE/ME</i>	4.9% (2.06)	0.9% (0.32)				12.7% (3.33)	−0.4% (−0.09)	−2.5% (−0.37)	8.0% (2.56)	1.2% (0.35)				14.0% (3.88)	0.6% (0.08)
<i>Gprof</i>	1.5% (0.70)		−2.1% (−0.81)			10.0% (2.84)	−1.6% (−0.34)	−1.8% (−0.31)	1.3% (0.47)		−1.1% (−0.39)			10.6% (3.60)	−0.7% (−0.12)
<i>Ag</i>	−3.8% (−2.06)			−2.9% (−1.29)				−3.3% (−0.94)	−8.3% (−2.31)			−3.3% (−0.90)			
<i>Size</i>	−4.4% (−1.89)				−2.7% (−1.08)			−2.8% (−0.97)	−6.4% (−2.19)				−2.5% (−0.85)		
Panel B: Equal-Weighted Portfolios															
Sorting	Decile portfolios based on included covariates								All 50 decile portfolios						
Variable	[1.1]	[1.2]	[1.3]	[1.4]	[1.5]	[1.6]	[1.7]	[1.8]	[2.1]	[2.2]	[2.3]	[2.4]	[2.5]	[2.6]	[2.7]
<i>Dur</i>	−9.1% (−4.40)	−8.6% (−3.36)	−9.6% (−4.02)	−8.9% (−4.09)	−9.7% (−4.52)		−8.4% (−1.84)	−8.4% (−1.78)	−11.7% (−4.91)	−10.5% (−4.12)	−12.0% (−4.56)	−10.3% (−4.54)	−11.5% (−4.72)		−8.9% (−1.91)
<i>BE/ME</i>	6.1% (2.53)	2.3% (0.80)				9.1% (3.31)	2.7% (0.54)	2.2% (0.42)	8.5% (3.28)	2.0% (0.73)				11.9% (4.18)	3.7% (0.73)
<i>Gprof</i>	4.0% (2.12)		−0.3% (−0.15)			7.5% (3.43)	1.2% (0.28)	2.0% (0.44)	3.8% (1.71)		−1.0% (−0.41)			8.9% (3.80)	1.8% (0.41)
<i>Ag</i>	−5.5% (−3.67)			−4.1% (−2.62)				−3.9% (−2.47)	−9.2% (−3.97)			−4.0% (−2.06)			
<i>Size</i>	−2.5% (−1.32)				−0.7% (−0.34)			−0.2% (−0.07)	−4.2% (−1.86)				−0.5% (−0.24)		

are still positive (although small and insignificant) and the low asset growth premium is still significant (3.9% with $t_{stat} = 2.47$).

None of the qualitative results described in the previous paragraphs changes in columns 2.1 to 2.7, where all 50 decile portfolios are kept in the panel data irrespective of the regression specification (note that specification 2.8 is not necessary since it would be identical to specification 1.8). Moreover, the Internet Appendix demonstrates that all results remain valid in firm-level Fama-MacBeth regressions that directly use the firm characteristics as covariates. The Internet Appendix also provides qualitatively similar results when state variables are removed from $s_{j,t}$ one at a time, with the biggest difference happening when gross profitability is removed from $s_{j,t}$ since it is an important predictor of future cash flows, but even then $Dur_{j,t}$ is strong and subsumes the value and profitability premia.

The overall conclusion is that the profitability and value premia exist only as a consequence of the endogenous correlation between duration, value, and profitability. Intuitively, if two firms have different growth opportunities, but the same profitability, the one with less growth opportunities (the value firm) will save/invest less given the lack of growth opportunities. Consequently, value

companies (controlling for profitability) tend to be short duration companies as they are expected to pay more of their resources to investors over the short term. Similarly, if two firms have different profitability, but similar growth opportunities, the more profitable one will have more resources to distribute to investors in the near future. Consequently, profitable companies (controlling for growth opportunities) also tend to be short duration firms.

4. Short duration premium and reinvestment risk

The empirical results in the previous section suggest that the short duration premium is likely an equilibrium outcome as opposed to an anomaly that concentrates in small stocks and quickly disappears. Moreover, the findings indicate that equity duration is linked to firm fundamentals and can help us better understand different phenomena in cross-sectional asset pricing. Consequently, it is important to understand what drives the short duration premium.

This section provides a novel explanation for why the short duration premium exists in financial markets. Section 4.1 builds on the ICAPM of Campbell (1993) to argue that long-term investors care about long-term wealth,

and thus price market risk (i.e., variation in current wealth) as well as reinvestment risk (i.e., variation in expected wealth growth). Section 4.2 empirically demonstrates that investors can only earn the short duration premium by being exposed to substantial reinvestment risk and this exposure is enough to explain the short-duration premium observed empirically. Finally, Section 4.3 shows that the short duration premium is much larger in periods in which earning the premium requires higher exposure to reinvestment risk.

4.1. An ICAPM with reinvestment risk

This Section details the ICAPM and explains how I measure the relevant risk factors.

4.1.1. The model

A long-term investor with relative risk aversion γ and initial wealth W_t chooses portfolio allocation to maximize expected utility of wealth H years ahead, $\mathbb{E}_t \left[\frac{1}{1-\gamma} \cdot W_{t+H}^{1-\gamma} \right]$. The investor can invest in a baseline asset with gross real return $R_{f,t}$ as well as in a set of risky assets with gross real return vector R_t to form the wealth portfolio, $R_{w,t}$. Wealth evolves according to $W_{t+1} = W_t \cdot R_{w,t+1}$, and the investor's allocation optimality conditions can be stated as (for any asset j)¹⁸

$$\mathbb{E}_{t-1} \left[R_{w,t}^{-\gamma} \cdot \left(\prod_{h=1}^{H-1} R_{w,t+h} \right)^{-(\gamma-1)} \cdot (R_{j,t} - R_{f,t}) \right] = 0. \quad (8)$$

Consequently, shocks to the long-term investor's log stochastic discount factor (SDF) are given by

$$\tilde{m}_t = -\gamma \cdot \tilde{r}_{w,t} - (\gamma - 1) \cdot N_{\mathbb{E},t}^{(H-1)} \quad (9)$$

so that the risk premium on any asset j relative to asset i can be written as¹⁹

$$\begin{aligned} \mathbb{E}_{t-1} [R_{j,t} - R_{i,t}] &= \gamma \cdot \text{Cov}_{t-1}(r_{j,t} - r_{i,t}, \tilde{r}_{w,t}) \\ &\quad + (\gamma - 1) \cdot \text{Cov}_{t-1}(r_{j,t} - r_{i,t}, N_{\mathbb{E},t}^{(H-1)}), \end{aligned} \quad (10)$$

↓

$$\begin{aligned} \mathbb{E}[R_{j,t} - R_{i,t}] &= \gamma \cdot \text{Cov}(r_{j,t} - r_{i,t}, \tilde{r}_{w,t}) \\ &\quad + (\gamma - 1) \cdot \text{Cov}(r_{j,t} - r_{i,t}, N_{\mathbb{E},t}^{(H-1)}). \end{aligned} \quad (11)$$

Intuitively, the long-term investor cares about wealth H years ahead so that shocks to current wealth, $\tilde{r}_{w,t} = \Delta w_t$, and news about expected wealth growth over the subsequent $H - 1$ years, $N_{\mathbb{E},t}^{(H-1)} = (\mathbb{E}_t - \mathbb{E}_{t-1}) \left[\sum_{h=1}^{H-1} r_{w,t+h} \right] =$

¹⁸ Letting π_t represent the wealth portfolio weights, the objective function can be written as $\mathbb{E}_t \left[\left(\prod_{h=2}^H R_{w,t+h} \right)^{1-\gamma} \cdot (R_{f,t+1} + \pi_t' (R_{t+1} - R_{f,t+1}))^{1-\gamma} \right]$ so that taking the derivative with respect to $\pi_{j,t}$ yields Eq. (8).

¹⁹ As in Campbell et al. (2018), this equation requires either joint normality for r_j , r_i , and m or the usual second order Taylor expansion for $E[e^{m_{j,t+1} + r_{j,t+1}}] = 1$. For notation convenience, I follow Campbell et al. (2018) and also use the approximation $\mathbb{E}_t[r_{j,t+1} - r_{i,t+1}] + \frac{1}{2}(\sigma_{j,t}^2 - \sigma_{i,t}^2) \approx \mathbb{E}_t[R_{j,t+1} - R_{i,t+1}]$. Otherwise, all results can be understood from the perspective of the relative “log risk premia,” $\mathbb{E}_t[r_{j,t+1} - r_{i,t+1}] + \frac{1}{2}(\sigma_{j,t}^2 - \sigma_{i,t}^2)$.

$(\mathbb{E}_t - \mathbb{E}_{t-1}) \left[\sum_{h=1}^{H-1} \Delta w_{t+h} \right]$, are both priced with $\tilde{r}_{w,t}$ capturing market risk and $N_{\mathbb{E},t}^{(H-1)}$ reinvestment risk.²⁰

To complete the model, I follow the ICAPM literature and assume the wealth portfolio is equal to the equity market portfolio; $r_{w,t} = r_{e,t}$.²¹ Following Binsbergen and Koijen (2010), I also assume (demeaned) equity expected log returns follow an autoregressive process of order one, AR(1):

$$\mathbb{E}_t r = \phi_r \cdot \mathbb{E}_{t-1} r + \tilde{\mathbb{E}}_t r. \quad (12)$$

This specification is convenient because it implies risk prices are fully characterized by relative risk aversion, γ , the investor's horizon, H , and the expected return persistence, ϕ_r :

$$\begin{aligned} \mathbb{E}[R_{j,t} - R_{i,t}] &= \gamma \cdot \text{Cov}(r_{j,t} - r_{i,t}, \tilde{r}_{e,t}) \\ &\quad + \lambda_{\mathbb{E},r} \cdot \text{Cov}(r_{j,t} - r_{i,t}, \tilde{\mathbb{E}}_t r), \end{aligned} \quad (13)$$

where $\lambda_{\mathbb{E},r} = (\gamma - 1) \cdot (1 - \phi_r^{H-1}) / (1 - \phi_r)$ so that the price of reinvestment risk increases with γ , H , and ϕ_r .

Importantly, the model nests the CAPM because $\lambda_{\mathbb{E},r} = 0$ if the investor has a one-period horizon ($H = 1$) or log utility ($\gamma = 1$). The model also reduces to the CAPM if there is no variation in expected returns ($\tilde{\mathbb{E}}_t r = 0$).

Similar to empirical tests of the CAPM, I keep the investment horizon, H , fixed over time. Despite this simplifying assumption, one can still view this framework as an equilibrium model (with a representative investor or limited stock market participation) in which the maximization of wealth over a given horizon represents a simplified way to model intertemporal choices of an infinitely lived agent.²²

4.1.2. Measuring risk factors

To measure the risk factors, $\tilde{r}_{e,t} = r_{e,t} - (\bar{r}_e + \mathbb{E}_{t-1} r)$ and $\tilde{\mathbb{E}}_t r = \mathbb{E}_t r - \phi_r \cdot \mathbb{E}_{t-1} r$, I need to estimate the expected return process. I assume (as in Binsbergen and Koijen, 2010) that (demeaned) expected log dividend growth is also an AR(1), $g_t = \phi_g \cdot g_{t-1} + \tilde{g}_t$, so that the log dividend price ratio, dp , follows an ARMA(2,1) that provides identification for $\max(\phi_r, \phi_g)$ and $\min(\phi_r, \phi_g)$.²³ A maximum

²⁰ The $\gamma > 1$ condition for $N_{\mathbb{E},t}^{(H-1)}$ to have a positive risk price is a consequence of two offsetting effects. An asset that comoves positively with reinvestment rates is desirable since it provides more capital to be invested when expected returns are high, allowing the investor to take advantage of the better investment opportunities. However, the asset also exposes investors to reinvestment risk. When $\gamma > 1$, the latter effect dominates so that the price of risk for news about expected returns is positive.

²¹ In this case, the expected return on wealth can only vary in equilibrium if volatility or risk prices also vary to keep the long-term investor satisfied with a fixed equity position. I do not take a stand on the source of variation in expected returns since it does not affect the unconditional Eq. (11), which is the base for my empirical analysis.

²² For instance, in the ICAPM of Campbell (1993), Eq. (13) still characterizes risk premia except that $\lambda_{\mathbb{E},r} = (\gamma - 1) \cdot \delta / (1 - \phi_r \cdot \delta)$, with δ representing a log-linearization constant. Moreover, if the intertemporal elasticity of substitution is 1, then δ is equal to the investor's time discount factor. As such, there is a direct mapping between H in the finite horizon model and the time discount factor in the infinite horizon model. Specifically, $\delta = (1 - \phi_r^{H-1}) / (1 - \phi_r)$.

²³ From Campbell and Shiller (1989), we have $dp_t - \bar{dp} = B_r \cdot \mathbb{E}_t r - B_g \cdot g_t$ if expected returns and dividend growth are AR(1) processes. As such,

likelihood estimation of this ARMA(2,1) process yields $\max(\phi_r, \phi_g) = 0.883$ and $\min(\phi_r, \phi_g) = 0.241$. Adding the identifying assumption that expected returns are more persistent than expected dividend growth, which is supported by the evidence in Binsbergen and Koijen (2010), I obtain $\phi_r = 0.880$ and $\phi_g = 0.238$. These estimates, which are entirely based on the dynamics of the dividend price ratio, are close to the $\phi_r = 0.932$ and $\phi_g = 0.354$ estimates in Binsbergen and Koijen (2010).

Given the ϕ_r estimate, I only need a proxy for $\mathbb{E}_t r$ to get the risk factors. Letting z_t represent demeaned aggregate state variables, I assume $\mathbb{E}_t r = b'z_t$ and estimate b by OLS.²⁴

In terms of data measurement, equity market returns and dividends are based on a value-weighted portfolio containing all common stocks available in the CRSP data set. I use six state variables in z_t (all measured in natural log units): dividend yield (dp), equity payout yield (poy), one-year Treasury yield (ty), term spread (TS), credit spread (CS), and value spread (VS).²⁵ All of these variables have been explored in the literature as important predictors equity returns.²⁶ The Internet Appendix provides results that exclude predictive variables from z_t one at a time to demonstrate the robustness of the ICAPM results to the specific aggregate state variables used.

dp is the sum of two AR(1) processes with persistence parameters ϕ_r and ϕ_g , which implies dp is a ARMA(2,1) with autoregressive parameters $\rho_1 = \phi_r + \phi_g$ and $\rho_2 = -\phi_r \cdot \phi_g$, and a moving average parameter that depends on the covariance structure of shocks (see Granger and Morris, 1976; Lütkepohl, 1984). Solving for ϕ_r and ϕ_g yields a quadratic system with solution $\max(\phi_r, \phi_g) = 0.5 \cdot (\rho_1 + \sqrt{\rho_1^2 + 4 \cdot \rho_2})$ and $\min(\phi_r, \phi_g) = 0.5 \cdot (\rho_1 - \sqrt{\rho_1^2 + 4 \cdot \rho_2})$.

²⁴ The OLS estimation of $b'z_t$ does not use information on dividend growth predictability. However, Campbell-Shiller decomposition implies $B_r \cdot \mathbb{E}_t r = (dp_t - \bar{dp}) + B_g \cdot g_t$ where $B_g = 1/(1 - \rho \cdot \phi_g)$ and $B_r = 1/(1 - \rho \cdot \phi_r)$, and Chen and Zhao (2009) argue that it is important to account for both return and dividend growth predictability when estimating ICAPM risk factors. As such, the Internet Appendix also provides results (consistent with the ones in the main text) using an alternative $\mathbb{E}_t r$ measure that accounts for both dividend growth and return predictability. Specifically, I estimate $b'z_t$ and $b'_g z_t$ by projecting returns and dividend growth onto z_t (using OLS) and set $\mathbb{E}_t r$ to the average of $b'z_t$ and $[(dp_t - \bar{dp}) + B_g \cdot b'_g z_t]/B_r$.

²⁵ The dividend yield is the log of aggregate dividends over a normalized index price. The equity payout yield is the log of (0.1 plus) aggregate net equity payout over market equity following Boudoukh et al. (2007). The term spread is the difference between the ten-year and one-year log Treasury yields. The credit spread is the difference between Moody's corporate BAA and AAA log yields. Following Campbell and Vuolteenaho (2004), the value spread is the difference between the log book-to-market ratios of the value and growth portfolios formed based on small stocks with an adjustment to account for within-year movements in market equity.

²⁶ Several papers use the dividend yield as a predictor for both dividend growth and stock returns, with theoretical justification provided by the valuation identity of Campbell and Shiller (1989). Modifications to this valuation identity can be used to motivate many additional valuation ratios as predictors for all three relevant variables. Following Boudoukh et al. (2007) and Larrain and Yogo (2008), I use the equity payout yield, The Treasury yield (Fama and Schwert, 1977; Fama, 1981), term spread (Campbell, 1987; Fama and French, 1989), and credit spread (Keim and Stambaugh, 1986) are classical interest rate and equity return predictors. Finally, Campbell and Vuolteenaho (2004), Campbell, Polk, and Vuolteenaho (2009), and Campbell et al. (2018) rely on the value spread as an important predictor of stock returns.

The dividend measurement used is based on the sum of annual dividends with no compounding to avoid introducing properties of returns into dividend growth (see Chen, 2009; Binsbergen and Koijen, 2010) and includes mergers and acquisitions paid in cash (as suggested by Allen and Michaely, 2003). Both aspects serve to make the dividend price ratio, which is an important state variable in my analysis, more stationary (consistent with Koijen and Nieuwerburgh, 2011; Sabbatucci, 2015).²⁷

Flow variables (such as dividend growth and returns) are deflated using the CPI. Moreover, to avoid seasonality issues, I use monthly observations of annual flows. The final data set used to estimate the risk factors is a multivariate time series of monthly observations in which flow variables have annual measurement; this data set extends from Dec-1952 to June-2018.²⁸ In the Internet Appendix, I provide a more detailed description of the data sources/measurement and also report the correlations between shocks to risk factors and state variables.

4.2. Results from the ICAPM estimation

I estimate the ICAPM to demonstrate that reinvestment risk is an empirically credible explanation for the existence of a short duration premium.

4.2.1. Risk exposures of duration portfolios

Table 4 provides the risk exposures of duration portfolios. The shocks and risk factors have annual measurement, so I compound the returns of duration portfolios from July of year t to June of year $t + 1$ to match with annual risk factors when calculating betas. Moreover, all betas in this section come from log excess returns to match the ICAPM pricing Eq. (13). I focus on value-weighted portfolios in my description.

Column $\bar{r}$ shows average excess returns calculated directly from annual returns, with results being very similar to the ones presented in Table 2 (short duration premium of 8.8% with $t_{\text{stat}} = 3.09$). Columns β_m and β_{Er} display the risk exposures relevant to the ICAPM pricing equation. Market betas strongly increase in equity duration ($\beta_m^{10-1} = 0.54$ with $t_{\text{stat}} = 4.98$), while reinvestment risk strongly decreases in equity duration ($\beta_{Er}^{10-1} = -1.89$ with $t_{\text{stat}} = -7.73$). Columns β_{dp} and β_{poy} show that the same reinvestment risk effect is observed if we proxy for shocks to expected returns using shocks to the dividend yield or the net payout yield, which is a common practice in the return predictability literature.²⁹

²⁷ The Internet Appendix further discusses these adjustments and provides similar results after measuring dividends without accounting for mergers and acquisitions paid in cash.

²⁸ The starting date is selected to strike a balance between a long sample period and consistency in the behavior of the state variables. In particular, the sample is based on the postwar period and starts after the Fed-Treasury Accord of 1951 that restored independence to the Fed, affecting monetary policy.

²⁹ Table 2 uses residuals of AR(1) processes to proxy for shocks to dp and poy . However, very similar results are obtained from other specifications, such as using residuals of a projection of dp_t or poy_t onto z_{t-1} . Since returns on the short duration strategy comove with shocks to expected returns no matter what expected return proxy we rely on, one

Table 4Risk exposures (β s) of duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from Eq. (7) (empirical details in Section 2). Monthly portfolio returns span the subsequent 12 months (from July/1973 to June/2018). The table reports average excess returns, $\bar{r}$, market risk, β_m , reinvestment risk, β_{Er} , as well as betas relative to dividend yield, β_{dp} , net payout yield, β_{poy} , dividend growth, $\beta_{\Delta d}$, and a downside market factor (defined as r_m when it is below one standard deviation from its mean), β_m^- . All statistics are based on annual returns (July/ t to June/ $t+1$), and betas are the regression slopes of log returns on shocks to risk factors. t_{stat} are in parentheses, and statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994).

Panel A: Value-weighted returns							
Dur decile	$\bar{r}$	β_m	β_{Er}	β_{dp}	β_{poy}	$\beta_{\Delta d}$	β_m^-
Short	13.3%	0.66	−0.41	−0.24	−0.17	0.14	0.04
2	11.9%	0.66	−0.45	−0.26	−0.21	0.02	0.11
3	13.2%	0.70	−0.49	−0.26	−0.19	0.07	0.97
4	12.6%	0.89	−0.75	−0.33	−0.29	0.04	0.69
5	11.7%	0.86	−1.21	−0.33	−0.41	0.01	0.81
6	9.4%	0.89	−1.31	−0.35	−0.40	−0.03	0.71
7	8.5%	0.75	−1.38	−0.40	−0.36	−0.17	0.85
8	7.6%	0.85	−1.36	−0.42	−0.37	−0.18	1.17
9	5.6%	0.90	−1.68	−0.44	−0.38	−0.14	0.99
Long	4.5%	1.20	−2.30	−0.56	−0.50	−0.12	2.47
L-S	−8.8%	0.54	−1.89	−0.32	−0.34	−0.26	2.43
(t_{L-S})	(−3.09)	(4.98)	(−7.73)	(−3.45)	(−2.32)	(−1.63)	(2.38)
Panel B: Equal-weighted returns							
Dur decile	$\bar{r}$	β_m	β_{Er}	β_{dp}	β_{poy}	$\beta_{\Delta d}$	β_m^-
Short	13.4%	0.66	−0.39	−0.33	−0.25	−0.03	−0.28
2	14.9%	0.76	−0.76	−0.34	−0.34	0.00	0.40
3	13.4%	0.77	−0.90	−0.32	−0.30	0.04	0.25
4	12.9%	0.80	−0.90	−0.37	−0.38	−0.01	0.55
5	11.5%	0.78	−0.90	−0.35	−0.31	−0.02	0.48
6	11.4%	0.81	−1.06	−0.37	−0.36	−0.02	0.91
7	9.6%	0.78	−1.04	−0.35	−0.37	−0.04	0.75
8	8.8%	0.87	−1.36	−0.42	−0.44	−0.09	0.71
9	6.1%	0.97	−1.73	−0.47	−0.45	−0.09	1.16
Long	3.0%	1.07	−2.57	−0.53	−0.63	−0.13	1.00
L-S	−10.4%	0.42	−2.18	−0.20	−0.38	−0.11	1.28
(t_{L-S})	(−3.42)	(2.50)	(−3.80)	(−1.79)	(−2.57)	(−0.58)	(0.94)

Intuitively, decreases in expected returns are associated with increases in cash flow present values through lower discount rates, and this effect is stronger for the present value of longer term cash flows. As such, β_{Er} decreases (becomes more negative) as equity duration increases. A simple way to understand this effect is to note that $Dur_t = -(\partial ME_t / \partial dr_t) / ME_t$, and thus equity duration reflects the proportional increase in equity value when the firm-specific discount rate decreases. If discount rates comove positively, then long duration firms will tend to go up in value by more than short duration firms when market expected returns decline.

Column $\beta_{\Delta d}$ shows the exposure of duration portfolios to shocks in the aggregate dividend growth; $\Delta d_t = \Delta d_t - b_g' z_{t-1}$ (with b_g estimated by OLS). There is a (statistically weak) decreasing pattern in dividend growth risk exposure ($\beta_{\Delta d}^{10-1} = -0.26$ with $t_{stat} = -1.63$), which is consistent with the prominent explanation for the short duration premium provided in Lettau and Wachter (2007, 2011). I later demonstrate that the time variation

in the short duration premium is consistent with the reinvestment risk channel but is inconsistent with this alternative risk-based explanation.

Since Cejnek and Randl (2016) relate the equity term structure to downside risk, column β_m^- shows the exposure of duration portfolios to market downside risk, measured in a way analogous to Lettau, Maggiori, and Weber (2014). Specifically, I estimate β_m^- using only observations in which $r_e < \bar{r}_e - \sigma_e$. The results show that long duration stocks have higher downside risk than short duration stocks, which indicates downside risk cannot explain the short duration premium.

4.2.2. ICAPM risk prices and pricing errors

While risk exposures are informative, combining them to study ICAPM-implied risk premia requires properly estimating risk prices, which is done in Table 5 based on the pricing Eq. (13). The first three columns use (value- and equal-weighted) equity duration portfolio spreads ($R_{Dur}^{(h)} - R_{Dur}^{(1)}$) as testing assets so that the slope of the equity term structure is the key “moment” the model attempts to match. The last three columns also require the model to perfectly match the equity premium so that the level of the equity term structure is also matched.

might be tempted to use R_{Dur}^{1-10} to predict equity returns. This is unlikely to be a productive path as R_{Dur}^{1-10} reflects shocks to expected returns, not the level of expected returns.

Table 5ICAPM estimation and pricing errors (α s) of duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from Eq. (7) (empirical details in Section 2). Monthly portfolio returns span the subsequent 12 months (from July/1973 to June/2018). The first three columns report risk prices and pricing errors (α s) for the estimation of Eq. (13) using as testing assets excess returns of (value- and equal-weighted) duration portfolios relative to the shortest duration portfolio ($R_{Dur}^{(h)} - R_{Dur}^{(1)}$). The last three columns further require the model to perfectly match the equity premium, $\mathbb{E}[R_e - R_f]$. The CAPM specification imposes $\gamma \geq 0$ and $\lambda_{\mathbb{E}r} = 0$, the ICAPM specification imposes $\gamma \geq 0$ and $0 \leq \lambda_{\mathbb{E}r}/(\gamma - 1) \leq (1 - \phi^{49})/(1 - \phi)$, and the ICAPM_U specification imposes no restriction on risk prices. Section 4.1 explains the construction of risk factors, and the Internet Appendix provides details on the model estimation. t_{stat} are in parentheses, and statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994).

	Matches $\mathbb{E}[R_{Dur}^{(h)} - R_{Dur}^{(1)}]$			Also imposes $\mathbb{E}[R_e - R_f]$		
	CAPM	ICAPM	ICAPM _U	CAPM	ICAPM	ICAPM _U
$\lambda_m = \gamma$	0.0 (0.00)	3.5 (1.90)	3.5 (1.16)	3.7 (-)	10.5 (6.60)	10.5 (6.37)
$\lambda_{\mathbb{E}r}$		16.4 (3.24)	16.4 (3.02)		24.9 (4.26)	24.9 (4.12)
implied H	1	13		1	4	
implied δ		0.97			0.79	
$\mathbb{E}[R_e - R_f]$	0.0%	-2.0%	-2.0%	7.3%	7.3%	7.3%
VW	α_{2-1}	-1.4%	-1.1%	-1.3%	-0.9%	-0.9%
	α_{3-1}	-0.1%	0.1%	-0.4%	-0.3%	-0.3%
	α_{4-1}	-0.7%	-0.3%	-2.6%	-2.8%	-2.8%
	α_{5-1}	-1.6%	2.0%	-3.2%	1.6%	1.6%
	α_{6-1}	-3.9%	0.1%	-5.7%	-0.3%	-0.3%
	α_{7-1}	-4.8%	0.7%	-5.5%	2.6%	2.6%
	α_{8-1}	-5.7%	-1.1%	-7.2%	-0.9%	-0.9%
	α_{9-1}	-7.7%	-1.5%	-9.6%	-0.9%	-0.9%
	α_{10-1}	-8.8%	-0.9%	-13.2%	-3.0%	-3.0%
	(t_{10-1}^{α})	(-3.00)	(-0.31)	(-4.49)	(-1.02)	(-1.02)
EW	α_{2-1}	1.5%	3.1%	0.7%	2.7%	2.7%
	α_{3-1}	-0.1%	2.3%	-1.0%	2.2%	2.2%
	α_{4-1}	-0.6%	1.6%	-1.8%	1.0%	1.0%
	α_{5-1}	-2.0%	0.3%	-3.0%	0.1%	0.1%
	α_{6-1}	-2.1%	1.0%	-3.3%	0.9%	0.9%
	α_{7-1}	-3.8%	-0.6%	-4.8%	-0.5%	-0.5%
	α_{8-1}	-4.7%	-0.1%	-6.4%	-0.2%	-0.2%
	α_{9-1}	-7.3%	-1.2%	-9.9%	-1.7%	-1.7%
	α_{10-1}	-10.4%	0.3%	-13.8%	1.0%	1.0%
	(t_{10-1}^{α})	(-3.45)	(0.09)	(-4.56)	(0.34)	(0.34)

The first specification, labeled CAPM, imposes $\gamma \geq 0$ and $\lambda_{\mathbb{E}r} = 0$ (i.e., $H = 1$) to reflect only market risk in the pricing equation. The second specification, labeled ICAPM, imposes $\gamma \geq 0$ and $0 \leq \lambda_{\mathbb{E}r}/(\gamma - 1) \leq (1 - \phi^{49})/(1 - \phi)$ (i.e., $1 \leq H \leq 50$) to reflect the ICAPM pricing equation with a wide range of potential investment horizons. The last specification, labeled ICAPM_U, imposes no restriction on $\lambda_m = \gamma$ and $\lambda_{\mathbb{E}r}$.³⁰ I focus on alphas of value-weighted portfolios in my description, but the alphas of equal-weighted portfolios are similar.

The CAPM clearly cannot capture the short duration premium. Since market betas increase in equity duration, $\gamma = 0$ when the estimation attempts to match only the slope of the equity term structure, leaving a large alpha for the short duration premium ($\alpha_{10-1} = -8.8\%$ with $t_{stat} = -3.00$). When required to match the equity premium of 7.3%, the CAPM estimation yields $\gamma = 3.7$ and

produces an even larger short duration premium alpha ($\alpha_{10-1} = -13.2\%$ with $t_{stat} = -4.49$).

In contrast to the CAPM, the ICAPM performs well in capturing the short duration premium. Specifically, the short duration premium alpha is -0.9% ($t_{stat} = -0.31$) when the estimation attempts to match only the slope of the equity term structure and -3.0% ($t_{stat} = -1.02$) when the equity premium is also imposed. The main difference between the two specifications is that risk aversion increases from 3.5 ($t_{stat} = 1.90$) to 10.5 ($t_{stat} = 6.60$) once we require the ICAPM to reproduce the equity premium. These results indicate that a moderate risk aversion is enough to produce the short duration premium, but capturing the equity and short duration premia jointly yields a relatively high risk aversion estimate (but not unreasonable as it is similar to the value used in the long-run risks literature). In terms of investor's horizon, ICAPM estimates imply a horizon of 13 years when not required to match the equity premium and 4 years when matching the equity premium (the respective annual time discount factors are 0.97 and 0.79 as per footnote 22). Interestingly, none of the ICAPM results change as we drop the model inequality restrictions on risk prices.

Fig. 4 provides a graphical illustration of the ICAPM alphas (for the estimation that imposes the equity premium). While average excess returns strongly decrease in

³⁰ All specifications are estimated using a pooled panel regression of returns on covariances while imposing the relevant equality and inequality restrictions. Standard errors are obtained using the Fama and MacBeth (1973) procedure. This entire estimation approach would be equivalent to Fama and MacBeth (1973) cross-sectional regressions of returns on covariances if there were no coefficient restrictions. A detailed description of the model estimation and inference is provided in the Internet Appendix.

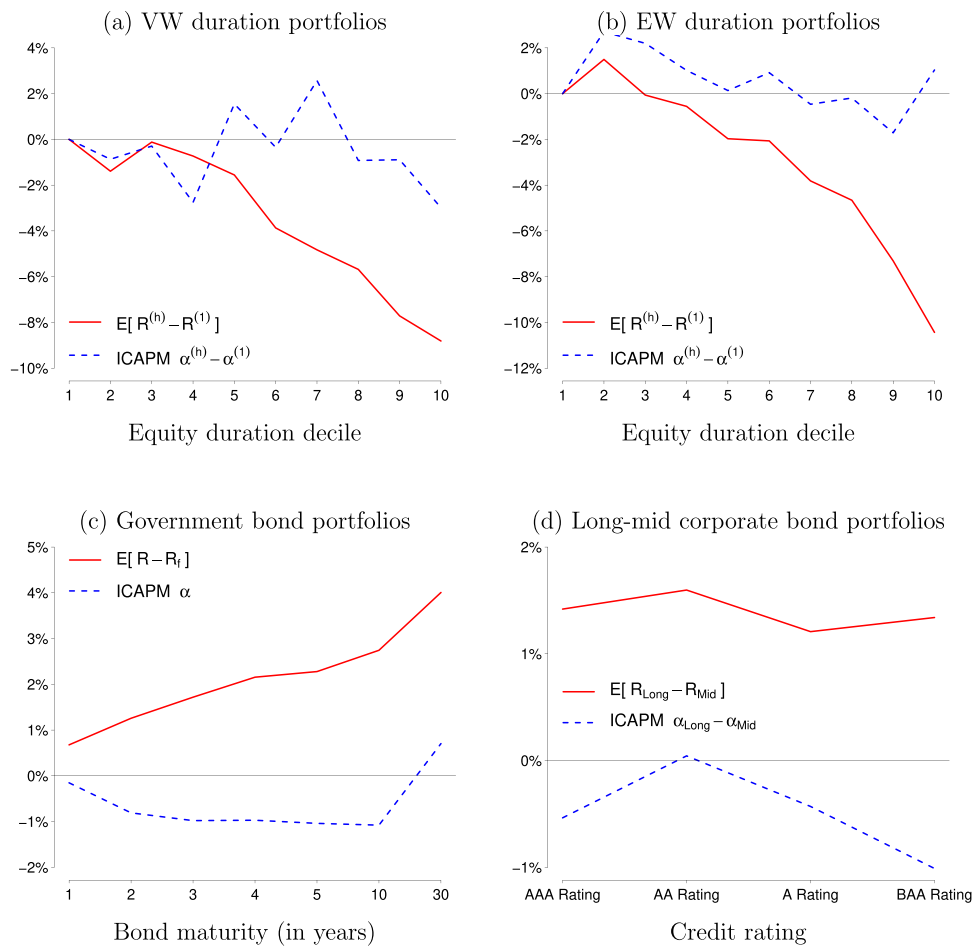


Fig. 4. Risk premia and ICAPM α s: equity duration portfolios and bond portfolios.

The graphs report average excess returns and ICAPM pricing errors (α s) for several long-short portfolios designed to capture the slopes of the equity and bond term structures. Risk prices come from the ICAPM estimation in column 5 of Table 5, which is based on the estimation of Eq. (13) using as testing assets excess returns of (value- and equal-weighted) duration portfolios relative to the shortest duration portfolio ($R_{Dur}^{(h)} - R_{Dur}^{(1)}$). The estimation further requires the model to perfectly match the equity premium, $\mathbb{E}[R_e - R_f]$, with risk prices that obey the ICAPM pricing restrictions $\gamma \geq 0$ and $0 \leq \lambda_{Er}/(\gamma - 1) \leq (1 - \phi^{49})/(1 - \phi)$.

equity duration, ICAPM α s display no pattern in value- or equal-weighted portfolios (Fig. 4(a) and (b)). Interestingly, despite the ICAPM estimation not using bond information, the model also goes a long way in capturing the fact that long duration (government and corporate) bonds have higher risk premia than short duration bonds (Fig. 4(c) and (d)).

These results demonstrate that the reinvestment risk mechanism captures the short duration premium in equities without producing a (counterfactual) short duration premium in bonds. In contrast to stocks, bond prices decrease when expected returns decline because nominal interest rates are negatively correlated with the equity premium (Fama and Schwert, 1977; Campbell, 1987; Ferson, 1989; Shanken, 1990; Brennan, 1997; Cederburg, 2019). Given their higher duration, longer term bonds are more exposed to this equity reinvestment risk and thus command higher risk premia.³¹

Overall, the findings in this section indicate that longer duration stocks are better hedges for reinvestment risk, and this effect is strong enough to induce a short duration premium of the magnitude observed in the data. Moreover, the same mechanism produces empirically credible bond risk premia, indicating that a unique channel can unify seemingly opposite term structure patterns observed in equities and bonds.

folios to reinvestment risk. Over my sample period, this negative correlation is strong with $\text{Cor}(\mathbb{E}r - ty, ty) = -0.57$ in shocks. Moreover, Gonçalves (2021) shows that the correlation between shocks to interest rates and the equity premium is consistently negative over time, in contrast to the correlation between realized equity and bond returns (see Baele, Bekaert, and Inghelbrecht, 2010; Campbell, Sunderam, and Viceira, 2017). In fact, the correlation is largely negative even over the period from 2000 to 2018, which is striking as 2000 is the year in which the correlation between realized equity and bond returns started to change sign (see Fig. 9.3 in Campbell, 2018).

³¹ The negative correlation between shocks to the equity premium and interest rates is a key force behind the exposure of bond port-

Table 6

Time variation in the short duration premium.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from Eq. (7) (empirical details in Section 2). Monthly portfolio returns span the subsequent 12 months (from July/1973 to June/2018). $\sigma(Dur)$ represents the cross-sectional standard deviation of $\log(Dur)$, with the logarithm transformation being used to decrease the large asymmetry in duration and the potential influence of outliers. The table reports results (average excess market returns as well as average returns and risk exposures of the long-short duration portfolio) separately for periods of low $\sigma(Dur)$ (lower 25% observations), moderate $\sigma(Dur)$, and high $\sigma(Dur)$ (highest 25% observations). Taking the low $\sigma(Dur)$ period as an example, I select the years (t) with the 25% lowest $\sigma(Dur)$, collect returns and risk factors from July of year $t + 1$ to June of year $t + 2$ (same time convention as the portfolio formation), and calculate average returns and covariances based on this subset of years. The betas reflect market risk, β_m , reinvestment risk, β_{Er} , as well as betas relative to dividend yield, β_{dp} , net payout yield, β_{poy} , and dividend growth, $\beta_{\Delta d}$. I provide t_{stat} for the tests comparing the low $\sigma(Dur)$ period with the respective (moderate or high) $\sigma(Dur)$ period. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994).

Panel A: Value-weighted returns							
$\sigma(Dur)$	$\bar{r}_e$	$\bar{r}_{10-1}$	β_m	β_{Er}	β_{dp}	β_{poy}	$\beta_{\Delta d}$
Low	5.6%	−3.1%	0.55	−1.02	−0.20	0.55	0.09
	(−)	(−)	(−)	(−)	(−)	(−)	(−)
Moderate	8.8%	−5.4%	0.61	−1.46	−0.50	−0.43	−0.73
	(0.58)	(−0.42)	(0.20)	(−0.37)	(−1.20)	(−1.68)	(−2.80)
High	5.9%	−20.3%	0.50	−2.56	−0.49	−0.88	−0.52
	(0.04)	(−2.38)	(−0.19)	(−1.33)	(−1.21)	(−2.47)	(−1.84)
Panel B: Equal-weighted returns							
$\sigma(Dur)$	$\bar{r}_e$	$\bar{r}_{10-1}$	β_m	β_{Er}	β_{dp}	β_{poy}	$\beta_{\Delta d}$
Low	5.6%	−3.6%	0.24	0.13	−0.01	0.45	0.04
	(−)	(−)	(−)	(−)	(−)	(−)	(−)
Moderate	8.8%	−7.4%	0.55	−1.77	−0.46	−0.47	−0.57
	(0.58)	(−0.69)	(0.87)	(−2.96)	(−1.87)	(−3.07)	(−1.89)
High	5.9%	−22.2%	0.49	−3.54	−0.28	−0.97	−0.24
	(0.04)	(−2.42)	(0.85)	(−3.20)	(−1.33)	(−3.62)	(−0.67)

4.3. Time-varying reinvestment risk and short duration premium

I now study the time variation in the reinvestment risk exposure of duration portfolios and its consequences for the short duration premium. I find that the short duration premium is substantially larger when earning the premium requires higher reinvestment risk exposure.

Equity duration, $Dur_t = -(\partial ME_t / \partial dr_t) / ME_t$, reflects the proportional increase in firm value when the firm specific discount rate decreases. As such, Dur directly reflects reinvestment risk exposure if there is a factor structure in discount rates. This connection suggests we should expect larger differences in reinvestment risk exposure across duration deciles when there is more cross-sectional variability in Dur . Therefore, taking the short duration strategy during these periods requires more reinvestment risk exposure, which should induce a higher short duration premium under the ICAPM (holding everything else fixed).

To test this prediction, I start by constructing a measure of the cross-sectional variability in duration, labeled $\sigma(Dur)$. Specifically, for each year t , I select all firms with fiscal year ending in December and measure the standard deviation of $\log(Dur_t)$ at that point in time. The December fiscal year is imposed to align all accounting information used to construct $\sigma(Dur)$, and the logarithm transformation is used to decrease the large asymmetry in duration and the potential influence of outliers.³²

Table 6 shows results (short duration premium and risk exposures) separately for periods of low $\sigma(Dur)$

(lowest 25% observations), moderate $\sigma(Dur)$, and high $\sigma(Dur)$ (highest 25% observations). Taking the low $\sigma(Dur)$ period as an example, I select the years (t) with the 25% lowest $\sigma(Dur)$, collect returns and risk factors from July of year $t + 1$ to June of year $t + 2$ (same time convention as the portfolio formation), and calculate average returns and covariances based on this subset of years. I provide t -statistics for the tests comparing the low $\sigma(Dur)$ period with the respective moderate or high $\sigma(Dur)$ period. I focus the discussion on value-weighted returns and point out when there are relevant differences relative to equal-weighted returns.

Column $\bar{r}_e$ reports average excess returns for the aggregate equity market and shows that the equity premium is similar for all three $\sigma(Dur)$ states. As such, the level of the equity term structure does not seem to vary with $\sigma(Dur)$. In contrast, as predicted by the link between $\sigma(Dur)$ and cross-sectional variation in reinvestment risk, $\sigma(Dur)$ is a strong predictor of the short duration premium. The premium is 3.1% following low $\sigma(Dur)$ periods and 20.3% following high $\sigma(Dur)$ periods ($t_{stat} = 2.38$ for the difference between the two periods).

The short duration strategy β_{Er} strongly varies with $\sigma(Dur)$ in a way consistent with the variation in the short duration premium. Specifically, $\beta_{Er} = -1.02$ during periods of low $\sigma(Dur)$ and $\beta_{Er} = -2.56$ during periods of high $\sigma(Dur)$. Despite the large shift in β_{Er} , the difference in β_{Er} across low and high $\sigma(Dur)$ periods is only significant in equal-weighted portfolios.

Fully testing the ICAPM in a conditional fashion is out of the scope of this paper as it requires taking a stand on how volatility and risk prices vary over time

³² Internet Appendix provides a plot of $\sigma(Dur)$ from 1973 to 2017.

(and imposing consistency between such variation and the expected return shocks). However, the overall results suggest that the variation in reinvestment risk is consistent with the variation in the short duration premium. Alternative explanations need to justify the large variation in the short duration premium across periods of low and high $\sigma(Dur)$, which is not trivial. For instance, explanations based on the conditional CAPM would fail as market beta variation would produce higher short duration premium during low $\sigma(Dur)$ periods, which is counterfactual (see column β_m). Similarly, exposure to cash flow shocks (as in Lettau and Wachter, 2007; Lettau and Wachter, 2011) predicts that the period of highest risk for the short duration strategy is the moderate $\sigma(Dur)$ period, when the short duration premium is only 5.4% (see column $\beta_{\Delta d}$). Further exploring time variation in the short duration premium is an interesting task for future research.

5. Exploring other equity duration measures

This section explores alternative equity duration measures. Section 5.1 demonstrates that equity duration measures that do not require a discount rate estimate, dr , yield short duration premia (and link to reinvestment risk) that are similar to my baseline results. Section 5.2 shows that a placebo duration measure that relies on the same $s_{j,t}$ as Dur but does not capture duration generates no spread in average returns or α s. Section 5.3 shows that my equity duration measure, Dur , improves upon the equity duration measure proposed in Dechow, Sloan, and Soliman (2004) on theoretical and empirical grounds.

5.1. Two alternative equity duration measures

This section shows that my main results are similar when I rely on two alternative equity duration measures that do not require a discount rate estimate.

5.1.1. Defining alternative measures of equity duration

The first alternative measure of equity duration I use keeps the nonlinearity of Dur without requiring a firm discount rate. Specifically, I define expected payback period (EPP) as the investment horizon that solves

$$\begin{aligned} ME_{j,t} &= \sum_{h=1}^{EPP_{j,t}} \mathbb{E}_t[PO_{j,t+h}] \\ &= \sum_{h=1}^{EPP_{j,t}} BE_{j,t} \cdot \left[e^{(\mathbf{1}_{CSprof} - \mathbf{1}_{BEg})' \Gamma^h s_{j,t} + \nu_1(h)} - 1 \right] \\ &\quad \cdot e^{\nu_2(h) + \mathbf{1}_{BEg}' (\sum_{\tau=1}^h \Gamma^\tau) \cdot s_{j,t}}, \end{aligned} \quad (14)$$

where the second equality follows from Eq. (5) and the VAR process in Eq. (4).

Finding EPP does not require a firm discount rate, and thus its value comes purely from cash flow timing. EPP answers the question, In expectation, how long will it take to recover the capital invested purely from cash flows? I use a root-finding algorithm for each firm/year separately on Eq. (14) to solve for $EPP_{j,t}$ given $s_{j,t}$ and the VAR parameter estimates.

The second alternative measure of equity duration keeps the concept of duration but ignores part of the inherent nonlinearity in Dur to eliminate the need for a discount rate to calculate duration. Specifically, I develop a log-linear approximation to firm value and use it to calculate a log-linear duration ($lDur = -\partial \log(ME)/\partial dr$), which does not depend on the firm discount rate. The derivation details are provided in the Internet Appendix, and the final expression is given by

$$\begin{aligned} lDur_{j,t} &= \sum_{h=1}^{\infty} \left\{ \prod_{\tau=0}^{h-1} \frac{e^{\mathbb{E}_t[mb_{j,t+\tau}]}}{e^{\mathbb{E}_t[mb_{j,t+\tau}]} + e^{\mathbb{E}_t[CSprof_{j,t+\tau} - BEg_{j,t+\tau}]} - 1} \right\} \\ &= \sum_{h=1}^{\infty} \left\{ \prod_{\tau=0}^{h-1} \frac{e^{\mathbf{1}'_{mb} \Gamma^\tau s_{j,t}}}{e^{\mathbf{1}'_{mb} \Gamma^\tau s_{j,t}} + e^{(\mathbf{1}_{CSprof} - \mathbf{1}_{BEg})' \Gamma^\tau s_{j,t}} - 1} \right\}, \end{aligned} \quad (15)$$

where mb is the log market-to-book ratio and the second equality follows from the VAR process in Eq. (4).

For both EPP and $lDur$, the (out-of-sample) VAR estimation is identical to the one used for Dur and is described in Section 2.

5.1.2. Empirical results using the alternative measures of equity duration

The Internet Appendix provides correlation matrices between the duration measures (and duration portfolio returns) I construct. Dur , EPP , and $lDur$ are strongly positively correlated, with the lowest correlation reported being 83% (between Dur and $lDur$).

Table 7 shows my main results after replacing Dur with EPP (Panel A) or $lDur$ (Panel B). I focus on value-weighted returns when describing results. The duration premium is very similar whether I use EPP (8.4% with $t_{stat} = 2.99$) or $lDur$ (9.4% with $t_{stat} = 3.32$) to capture duration. For both measures, the premium remains strong over a period of five years and when I focus only on large companies. Moreover, α s relative to the ICAPM presented in Section 4.1 are small and insignificant (risk premia estimates are provided in the Internet Appendix). Finally, in both cases, the favorable ICAPM performance is driven by reinvestment risk, with the short duration strategy being strongly exposed to declines in equity expected returns.

Overall, the results indicate that the short duration premium and its link to reinvestment risk remain similar after replacing Dur by two alternative measures of cash flow duration (EPP and $lDur$) that do not require a discount rate estimate.

5.2. A placebo duration measure

Dur (as well as EPP and $lDur$) is a function of the firm-level characteristics in $s_{j,t}$. One might conjecture that the large spread in expected returns obtained from portfolios sorted on Dur is not related to duration itself but instead is a result of duration being a function of characteristics that proxy for expected returns. While this hypothesis is plausible, this section demonstrates it does not explain my results.

On a theoretical level, a characteristic in $s_{j,t}$ only affects Dur through its ability to predict relative cash flows going forward. For instance, if gross profitability did

Table 7

Risk and average returns of alternative duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on *EPP* and *ILDur* (details in Section 5.1), and portfolio returns are measured monthly. The table reports annualized average returns, $\bar{r}$ ($\times 12$), ICAPM pricing errors based on Eq. (13), α_{ICAPM} , market risk, β_m , and reinvestment risk, β_{Er} . $\bar{r}_{t \rightarrow t+h}$ indicates the portfolio is held for h years, and $\bar{r}_{t \rightarrow t+1}^{large}$ indicates the portfolio is based on firms with market equity above the 80% NYSE quantile. Statistics related to the ICAPM are based on annual returns (July/ t to June/ $t+1$), and betas are the regression slopes of log returns on shocks to risk factor. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West 1987; 1994) with t_{stat} in parentheses.

Panel A: Duration = EPR_t													
Duration	Value-weighted portfolios						Duration	Equal-weighted portfolios					
decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	α_{ICAPM}	β_m	β_{Er}	decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	α_{ICAPM}	β_m	β_{Er}
Short	12.7%	11.6%	10.4%	0.0%	0.66	−0.36	Short	13.0%	12.7%	10.8%	0.0%	0.66	−0.31
2	11.9%	10.9%	11.1%	3.0%	0.62	−0.40	2	14.0%	13.3%	10.9%	0.2%	0.75	−0.76
3	11.9%	11.1%	9.7%	2.4%	0.71	−0.46	3	13.1%	12.7%	10.3%	−1.0%	0.74	−0.80
4	11.4%	10.5%	9.6%	0.7%	0.92	−0.64	4	12.3%	12.3%	10.2%	−4.1%	0.82	−0.88
5	11.2%	9.6%	8.9%	−0.4%	0.83	−0.82	5	11.2%	11.8%	9.2%	−1.2%	0.79	−0.84
6	9.1%	9.1%	5.2%	1.4%	0.83	−1.18	6	11.1%	11.5%	7.5%	−0.5%	0.79	−1.07
7	9.1%	8.0%	7.2%	−2.2%	0.79	−1.00	7	9.9%	10.4%	7.7%	−1.1%	0.81	−0.86
8	7.9%	7.3%	6.8%	0.4%	0.84	−1.26	8	9.0%	9.6%	7.9%	−1.4%	0.85	−1.38
9	6.3%	6.4%	3.8%	−1.0%	0.91	−1.59	9	6.2%	8.1%	4.8%	−1.7%	0.95	−1.84
Long	4.2%	4.8%	3.0%	1.2%	1.20	−2.73	Long	3.7%	5.8%	4.9%	−0.3%	1.11	−2.84
L-S	−8.4%	−6.9%	−7.3%	1.2%	0.54	−2.37	L-S	−9.3%	−6.9%	−5.9%	−0.3%	0.45	−2.53
(t_{L-S})	(−2.99)	(−3.43)	(−2.23)	(0.42)	(4.29)	(−5.95)	(t_{L-S})	(−3.23)	(−2.83)	(−2.18)	(−0.10)	(2.66)	(−4.02)

Panel B: Duration = $ILDur_t$													
Duration	Value-weighted portfolios						Duration	Equal-weighted portfolios					
decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	α_{ICAPM}	β_m	β_{Er}	decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	α_{ICAPM}	β_m	β_{Er}
Short	12.5%	11.7%	10.2%	0.0%	0.70	−0.27	Short	13.8%	13.1%	11.2%	0.0%	0.66	−0.28
2	12.2%	10.7%	9.6%	0.3%	0.61	−0.30	2	13.2%	12.7%	10.3%	1.5%	0.76	−0.63
3	11.5%	10.4%	8.4%	−1.6%	0.73	−0.51	3	11.7%	12.3%	9.9%	0.3%	0.76	−0.64
4	11.3%	9.9%	9.0%	−0.4%	0.89	−0.47	4	11.7%	12.1%	9.2%	−3.4%	0.83	−0.95
5	9.6%	9.2%	7.8%	−0.2%	0.90	−0.94	5	12.1%	11.8%	8.5%	−1.6%	0.84	−0.95
6	9.4%	8.8%	7.6%	0.1%	0.78	−0.82	6	11.6%	11.4%	8.3%	−0.6%	0.79	−0.94
7	9.6%	8.7%	7.6%	−0.2%	0.76	−0.81	7	9.7%	10.4%	8.4%	0.1%	0.86	−1.34
8	7.5%	7.3%	5.6%	−0.3%	0.78	−1.12	8	8.7%	9.6%	7.0%	−0.2%	0.89	−1.53
9	7.5%	6.8%	5.3%	0.6%	0.95	−1.82	9	7.7%	8.8%	7.4%	2.6%	0.87	−1.73
Long	3.1%	4.5%	2.5%	0.1%	1.22	−2.52	Long	3.2%	6.0%	3.9%	−1.9%	1.00	−2.59
L-S	−9.4%	−7.2%	−7.7%	0.1%	0.52	−2.25	L-S	−10.6%	−7.2%	−7.4%	−1.9%	0.35	−2.31
(t_{L-S})	(−3.32)	(−3.04)	(−2.57)	(0.04)	(4.14)	(−4.37)	(t_{L-S})	(−3.58)	(−2.57)	(−2.55)	(−0.66)	(2.17)	(−3.47)

not predict cash flows at any horizon, then the Γ matrix would be such that *Dur* would not be a function of gross profitability even though gross profitability is included in $s_{j,t}$. *Dur* combines the characteristics in a nonlinear way to capture equity duration, not expected returns. The combination captures expected returns indirectly simply because equity duration is related to expected returns. This approach contrasts with Lewellen (2015), for example, as his combination of characteristics is explicitly created to capture expected returns.

On the other hand, it is possible that even arbitrary functions of $s_{j,t}$ have a tendency to predict returns (in one direction or the other) simply because the variables in $s_{j,t}$ tend to predict returns and are correlated with one another.

To empirically explore this issue, I create a simple placebo duration variable, *pDur*, that uses the same state variables as *Dur* but combines them in a way that is unrelated to equity duration. Specifically, $pDur_{j,t} = b'_p s_{j,t}$, where b_p is the vector of coefficients from a cross-sectional regression of firm name alphabetical ranks on $s_{j,t}$, $NAME\ RANK_t = b'_p s_{j,t} + \varepsilon_{j,t}$.

Table 8 replicates Table 2 after replacing *Dur* with *pDur*. As it is clear from the table, *pDur* has no relation to expected returns or α s despite being a function of $s_{j,t}$. This result confirms the point that my findings are not simply due to including in $s_{j,t}$ variables that correlate with expected returns. Quite the contrary, I argue that these variables predict returns precisely because they correlate with duration as demonstrated in Section 3.4.

5.3. Comparison with Dechow, Sloan, and Soliman (2004)

In an important contribution, Dechow, Sloan, and Soliman (2004) (henceforth DSS) provide a measure of equity duration and study some of its properties. This section contrasts DSS's equity duration measure (*DSS Dur*) with the one I develop in this paper (*Dur*) both on theoretical and empirical grounds to demonstrate that *Dur* improves upon *DSS Dur* on several dimensions.

5.3.1. Theoretical comparison between *Dur* and *DSS Dur*

Duration can be generally defined as in Eq. (1). I use a VAR model to get expressions for the cash flow ex-

Table 8Performance of duration portfolios based on a placebo *Dur*.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on a placebo duration variable, $pDur_t$, that uses the same state variables as *Dur* but combines them in a way that is unrelated to equity duration. Specifically, $pDur_t = b'_p s_t$, where b_p is the vector of coefficients from a cross-sectional regression of firm name alphabetical ranks on s_t . Panel A shows annualized average excess returns ($\times 12$) and Sharpe ratios ($\times \sqrt{12}$). $\bar{r}_{t+j \rightarrow t+h}$ indicates the portfolio is implemented j years after the *Dur* measurement and is held for $h-j$ years. Moreover, $\bar{r}_{t \rightarrow t+1}^{large}$ indicates the portfolio is based on firms with market equity above the 80% NYSE quantile, and $\bar{r}_{t \rightarrow t+1}^{dist adj}$ indicates returns are adjusted for delistings. Panel B reports β s and annualized α s ($\times 12$) from factor regressions. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994) with t_{stat} in parentheses and p -values in brackets.

Panel A: Average returns and Sharpe ratios													
Value-weighted portfolios							Equal-weighted portfolios						
Duration							Duration						
decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{large}$	$\bar{r}_{t \rightarrow t+1}^{dist adj}$	$\bar{r}/\sigma$	decile	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{large}$	$\bar{r}_{t \rightarrow t+1}^{dist adj}$	$\bar{r}/\sigma$
Short	5.9%	6.7%	8.2%	5.2%	5.9%	0.31	Short	7.8%	9.1%	10.7%	6.6%	7.7%	0.33
2	6.6%	7.1%	8.4%	6.1%	6.6%	0.39	2	10.2%	10.8%	10.5%	8.5%	10.2%	0.47
3	8.4%	7.6%	7.6%	7.7%	8.4%	0.48	3	10.1%	11.1%	12.1%	8.5%	10.1%	0.51
4	8.2%	8.1%	8.4%	7.3%	8.2%	0.52	4	10.6%	11.2%	11.6%	7.3%	10.5%	0.55
5	8.6%	8.1%	8.9%	8.2%	8.6%	0.50	5	11.1%	11.3%	11.5%	8.3%	11.1%	0.57
6	7.0%	6.9%	7.7%	7.0%	7.1%	0.42	6	11.4%	11.3%	10.5%	9.8%	11.4%	0.59
7	8.9%	7.7%	7.3%	5.5%	8.9%	0.52	7	10.7%	11.0%	10.8%	7.8%	10.6%	0.55
8	8.5%	8.7%	9.1%	9.0%	8.5%	0.51	8	10.9%	11.3%	11.2%	9.5%	10.9%	0.56
9	7.9%	8.0%	7.8%	7.0%	7.9%	0.44	9	10.8%	10.9%	11.3%	9.0%	10.7%	0.53
Long	8.7%	8.0%	9.1%	9.4%	8.8%	0.43	Long	9.9%	10.2%	10.0%	8.9%	9.7%	0.43
L-S (t_{L-S})	2.8% (1.36)	1.2% (0.81)	0.8% (0.42)	4.2% (1.94)	2.9% (1.37)	0.22 [0.26]	L-S (t_{L-S})	2.1% (0.96)	1.1% (0.64)	−0.7% (−0.34)	2.4% (1.37)	2.0% (0.90)	0.17 [0.24]
Panel B: Risk-adjusted performance based on factor models													
Duration	CAPM		Fama and French (2015) 5-factors						Hou, Xue, and Zhang (2015) q-factors				
	α_{CAPM}	β_{MKT}	α_{FF}	β_{MKT}	β_{SMB}	β_{HML}	β_{CMA}	β_{RMW}	α_q	β_{MKT}	β_{SIZE}	β_{INV}	β_{ROE}
Value-weighted portfolios													
Short	−2.0%	1.11	−0.5%	1.05	0.04	−0.28	−0.06	0.03	−0.5%	1.07	0.01	−0.34	0.09
2	−0.6%	1.00	−1.0%	1.01	−0.02	−0.06	−0.03	0.19	−1.0%	1.01	−0.04	−0.08	0.17
3	0.8%	1.04	0.6%	1.05	−0.03	−0.18	0.18	0.06	0.6%	1.05	−0.02	−0.04	0.10
4	1.6%	0.92	−0.2%	0.97	0.02	−0.06	0.22	0.24	0.5%	0.95	−0.03	0.09	0.13
5	1.4%	1.00	−0.1%	1.05	0.01	−0.14	0.26	0.21	0.9%	1.03	−0.06	0.07	0.07
6	0.0%	0.98	−0.9%	1.00	0.03	−0.06	0.04	0.21	0.1%	0.99	−0.05	−0.01	0.06
7	1.4%	1.00	−0.4%	1.03	0.17	0.01	0.15	0.15	0.1%	1.01	0.15	0.10	0.08
8	1.3%	0.96	−1.3%	1.02	0.10	−0.06	0.27	0.29	−1.0%	1.00	0.06	0.20	0.18
9	0.1%	1.04	0.5%	1.01	0.07	0.02	−0.09	−0.04	1.5%	1.01	0.01	−0.12	−0.07
Long	0.1%	1.17	−0.6%	1.13	0.31	0.10	−0.12	0.05	1.0%	1.13	0.22	−0.04	−0.13
L-S (t_{L-S})	2.1% (0.94)	0.06 (0.84)	−0.1% (−0.04)	0.08 (1.39)	0.27 (3.73)	0.38 (3.42)	−0.06 (−0.42)	0.01 (0.12)	1.5% (0.55)	0.05 (0.75)	0.21 (2.92)	0.30 (2.00)	−0.22 (−1.40)
Equal-weighted portfolios													
Short	−2.2%	1.35	−0.4%	1.15	0.69	−0.29	0.00	−0.39	1.0%	1.16	0.66	−0.41	−0.34
2	1.1%	1.24	0.7%	1.12	0.64	0.01	−0.11	−0.09	2.5%	1.11	0.55	−0.17	−0.24
3	1.6%	1.14	0.1%	1.07	0.57	0.05	−0.04	0.07	1.6%	1.06	0.48	−0.02	−0.13
4	2.2%	1.12	−0.4%	1.08	0.58	0.14	0.02	0.14	1.2%	1.05	0.49	0.12	−0.09
5	2.7%	1.14	0.0%	1.11	0.56	0.12	0.02	0.18	1.6%	1.08	0.46	0.11	−0.07
6	3.2%	1.11	−0.2%	1.11	0.57	0.15	0.11	0.23	1.3%	1.07	0.47	0.23	−0.05
7	2.4%	1.12	−0.9%	1.10	0.62	0.23	0.00	0.19	0.7%	1.07	0.52	0.21	−0.08
8	2.6%	1.13	−1.0%	1.11	0.65	0.25	0.00	0.23	0.5%	1.08	0.54	0.24	−0.06
9	2.4%	1.14	−2.1%	1.14	0.67	0.25	0.06	0.34	−0.2%	1.11	0.53	0.36	−0.07
Long	0.6%	1.25	−2.8%	1.19	0.80	0.28	−0.07	0.15	−0.6%	1.17	0.65	0.23	−0.21
L-S (t_{L-S})	2.8% (1.20)	−0.10 (−0.99)	−2.4% (−1.01)	0.04 (0.71)	0.11 (1.49)	0.57 (4.98)	−0.07 (−0.56)	0.54 (5.18)	−1.6% (−0.53)	0.01 (0.13)	0.00 (−0.01)	0.64 (2.56)	0.13 (0.62)

pectations, $\mathbb{E}_t[CF_{t+h}]$, in Eq. (1) and calculate duration. In contrast, DSS make several assumptions to deal with the same cash flow issue. First, DSS split Eq. (1) into the cash flows until (an arbitrary) time $t+H$ and the cash flows after that:

$$DSS Dur_{j,t} = \sum_{h=1}^H w_{j,t}^{(h)} \cdot h + \sum_{h=H+1}^{\infty} w_{j,t}^{(h)} \cdot h$$

$$= \sum_{h=1}^H w_{j,t}^{(h)} \cdot h + \left(1 - \sum_{h=1}^H w_{j,t}^{(h)}\right) \cdot \left(H + \frac{e^{dr}}{e^{dr} - 1}\right), \quad (16)$$

where $w_{t,h} = (\mathbb{E}_t[CF_{j,t+h}] \cdot e^{-h \cdot dr}) / ME_t$ and $ME_{j,t,T} = \sum_{h=1}^H \mathbb{E}_t[CF_{j,t+h}] \cdot e^{-h \cdot dr}$.

The second equality follows from the assumption that cash flows after time $t+H$ are fixed, which makes the sec-

ond component of the duration equation a level perpetuity with value $ME_{j,t} - ME_{j,t,T}$ and duration $H + (e^{dr}/e^{dr} - 1)$ as of time t .

To get expectations for cash flows from $t + 1$ to $t + H$, DSS assume clean surplus accounting holds such that clean surplus earnings can be replaced by accounting earnings (measured as income before extraordinary items) to get $CF_{j,t} = E_{j,t} - \Delta BE_{j,t}$, which implies

$$\begin{aligned}\mathbb{E}_t[CF_{j,t+h}] &= \mathbb{E}_t\left[BE_{j,t+h-1} \left(\frac{E_{j,t+h}}{BE_{j,t+h-1}} - \frac{\Delta BE_{j,t+h}}{BE_{j,t+h-1}}\right)\right] \\ &= \mathbb{E}_t[BE_{j,t+h-1} \cdot (ROE_{j,t+h} - BEG_{j,t+h})].\end{aligned}\quad (17)$$

DSS then assume book equity growth ($BEG_{j,t}$) and return on equity ($ROE_{j,t}$) follow first order univariate autoregressive processes and that sales growth ($YG_{j,t}$) is the right predictor for book equity growth so that $\mathbb{E}_t[BEG_{j,t+h}] = \bar{G} + \rho_G^h \cdot (YG_{j,t} - \bar{G})$, $\mathbb{E}_t[BE_{j,t+h-1}] = BE_{j,t} \cdot [1 + \bar{G} + \rho_G^{h-1} \cdot (YG_{j,t} - \bar{G})]$, and $\mathbb{E}_t[ROE_{j,t+h}] = \bar{ROE} + \rho_{ROE}^h \cdot (ROE_{j,t} - \bar{ROE})$. Consequently, DSS *Dur* is based on the following cash flow forecast:

$$\begin{aligned}\mathbb{E}_t[CF_{j,t+h}] &= BE_{j,t} \cdot [1 + \bar{G} + \rho_G^{h-1} \cdot (YG_{j,t} - \bar{G})] \\ &\quad \times [\bar{ROE} + \rho_{ROE}^h \cdot (ROE_{j,t} - \bar{ROE}) - \bar{G} - \rho_G^h \cdot (YG_{j,t} - \bar{G})],\end{aligned}\quad (18)$$

which, together with an exogenously imposed dr , allows DSS to recover $w_{j,t}^{(h)}$ for $h = 1, \dots, H$ and calculate DSS *Dur*.

While DSS *Dur* is a very important first step in measuring equity duration, a few points make it clear that *Dur* improves upon DSS *Dur* from a theoretical perspective.

First, the assumption of a level perpetuity is strong. It assumes cash flows no longer grow after year $t + H$. This induces a strong downward bias in DSS *Dur*. For instance, Weber (2018) relies on DSS *Dur* and reports an average duration of 6 years for the low duration decile and an average duration of 24 years for the high duration decile. Using the simple log-linear approximation explained in footnote 11 (which only requires dividend yield to obtain duration), the aggregate duration is between 38.1 and 48.6 years, which is in line with estimates based on *Dur* (see Table 1) but much higher than the DSS *Dur* of the high duration portfolio in Weber (2018).

Second, while *Dur* estimates expected cash flows directly from the dynamics of net payouts, which are cash flows to stockholders, DSS *Dur* does not use cash flow information to measure the expectation of long-term cash flows. Specifically, the assumption that the level perpetuity to be received at time $t + H$ has value $ME_{j,t} - ME_{j,t,T}$ at time t implies $\mathbb{E}_t[CF_{j,\infty}] = (ME_{j,t} - ME_{j,t,T}) \cdot (1 - e^{-dr}) \cdot e^{-(H+1) \cdot dr}$, where $CF_{j,\infty}$ is the (fixed, but unknown) cash flow to be received every year from $t + H + 1$ to $t + \infty$.³³ Since dr is exogenously specified, the expectation of long-term cash flows to be paid by the firm is not based on the dynamics of cash flows (or earnings) but instead on the current market equity of the firm, $ME_{j,t}$,

in excess of the present value of short-term cash flows, $ME_{j,t,T}$.

Third, even short-term expected cash flows in DSS *Dur* are not estimated from the dynamics of cash flows but instead from the dynamics of earnings. By replacing clean surplus earnings with accounting earnings, DSS measure equity duration without ever using firm payout information. In contrast, *Dur* constructs clean surplus earnings from equity payouts, effectively relying on cash flows to estimate duration.

Fourth, the assumptions imposed to derive DSS *Dur* are internally inconsistent. To move from Eq. (17) to Eq. (18), DSS (implicitly) assume that $\mathbb{E}_t[BE_{j,t+h-1} \cdot (ROE_{j,t+h} - BEG_{j,t+h})] = \mathbb{E}_t[BE_{j,t+h-1}] \cdot \mathbb{E}_t[ROE_{j,t+h} - BEG_{j,t+h}]$, which requires lagged growth to be uncorrelated with current growth and ROE. The assumption that BEG follows a univariate process that depends on YG , which is autocorrelated, implies the zero correlation assumption cannot hold, and thus the assumptions are internally inconsistent.

Finally, the assumption of univariate autoregressive processes for ROE and growth can be viewed as a restriction on the more general assumption I make about a VAR process including these variables (in log units). As such, *Dur* can be seen as a generalization of DSS *Dur*.

Overall, DSS *Dur* is an important first step in measuring equity duration, but *Dur* improves upon it on several theoretical aspects.

5.3.2. Empirical comparison between *Dur* and DSS *Dur*

Fig. 5 replicates Fig. 1 after replacing *Dur* with DSS *Dur*. Fig. 5(a) and (b) show that DSS *Dur* is, in fact, a measure of equity duration since firms in the short duration portfolio tend to pay more cash flows to investors (as a fraction of initial investment) over the short term than firms in the long duration portfolio. However, the pattern is not as strong as with *Dur*, with Fig. 5(a) indicating decile five pays as much cash flows over the short term as decile one. Similarly, Fig. 5(c) and (d) indicate that long duration firms are more exposed to the dividend term structure, but the pattern is far from monotone and concentrates in the longest duration deciles. The conclusion is that DSS *Dur* captures equity duration but not as well as *Dur*.

Table 9 replicates Table 2 after replacing *Dur* with DSS *Dur*. The short duration premium is smaller (6.1% and 7.8% in value- and equal-weighted portfolios) and delivers no α relative to Fama and French (2015)'s five-factor model or Hou, Xue, and Zhang (2015)'s q-factor model whether we focus on value- or equal-weighted returns. This result, which is quantitatively similar to the findings in Hou, Xue, and Zhang (2015, 2019), indicates that *Dur* also improves upon DSS *Dur* in terms of providing a higher spread in average returns that is not fully captured by standard factor models.

5.3.3. Mispricing and reinvestment risk

Weber (2018) finds a much larger DSS *Dur* short duration premium that is not fully captured by the Fama and French (2015)'s five-factor model and argues that market participants are overly optimistic

³³ To see this result, note that the value of the level perpetuity is given by $\sum_{h=H+1}^{\infty} \mathbb{E}_t[CF_{j,\infty}] \cdot e^{-h \cdot dr} = \mathbb{E}_t[CF_{j,\infty}] \cdot e^{-(H+1) \cdot dr} / (1 - e^{-dr})$, which is assumed to be equal to $ME_{j,t} - ME_{j,t,T}$, yielding the given implied cash flow expectation.

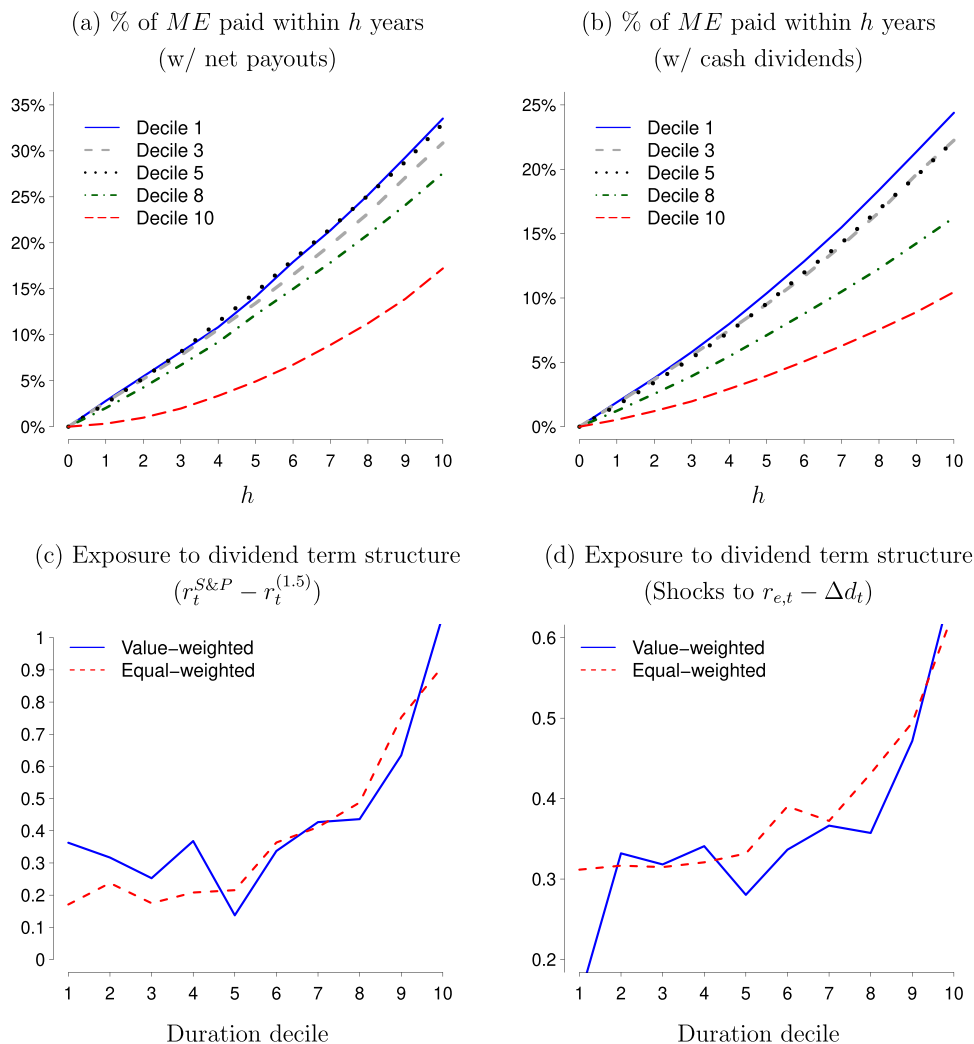


Fig. 5. Validating [Dechow, Sloan, and Soliman \(2004\)](#) duration portfolios.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on Dur , which is measured from [Eq. \(16\)](#) (empirical details in [Section 5.3](#)). Graph (a) shows the cumulative fraction of market equity that firms in each duration decile pay (in net payouts) over the ten years following the duration measurement. Graph (b) repeats this analysis after replacing net payouts with cash dividends. Graphs (c) and (d) show the exposures (regression slopes) of duration portfolios (value-weighted and equal-weighted) to the dividend term structure, measured as returns on a long short portfolio that buys a long duration asset (an equity index) and sells a short duration asset (a short-term dividend claim). Graph (c) uses annual log returns on the S&P 500 as the long-duration asset and annual log returns on a S&P 500 dividend claim (with approximately 1.5 year maturity) as the short duration asset (first annual return is in June/1997). Graph (d) provides results using (shocks to) annual log returns on the CRSP equity portfolio as the long duration asset and (shocks to) annual log dividend growth on the same portfolio as the short duration asset. All necessary details are provided in [Section 3.2](#).

about the prospects of long duration companies, inducing overvaluation for these companies, which leads to poor returns. The results in [Table 9](#) do not represent a failure to replicate [Weber \(2018\)](#)'s finding. Specifically, the Internet Appendix replicates the results in [Weber \(2018\)](#) after matching his empirical specification (dropping the use of NYSE breakpoints, keeping microcaps in equal-weighted portfolios, and winsorizing returns at 1% and 99%).³⁴

Overall, equity duration correlates with both mispricing and reinvestment risk. Given [Weber \(2018\)](#)'s goal of

studying the effect of mispricing on the short duration premium, his empirical decisions put the spotlight on small firms. In contrast, my goal is to study the short duration premium that exists even in the absence of mispricing, which motivates my empirical decision to minimize the effect of mispricing by focusing on relatively large firms (through NYSE breakpoints in value-weighted portfolio or the removal of microcaps in equal-weighted portfolios).

The natural interpretation of the overall evidence (once we account for both papers) is that reinvestment risk and mispricing are complementary channels that help capture the short duration premium. Specifically, reinvestment risk justifies the existence of a short duration premium in the

³⁴ The Internet Appendix further shows that the Dur short duration premium is also stronger after dropping the use of NYSE breakpoints, keeping microcaps in equal-weighted portfolios, and winsorizing returns.

Table 9Performance of duration portfolios based on *DSS Dur*.

Equity duration portfolios are formed every June (1973 to 2017) from deciles based on *DSS Dur*, which is measured from Eq. (16) (empirical details in Section 5.3), and portfolio returns are measured monthly. Panel A shows annualized average excess returns ($\times 12$) and Sharpe ratios ($\times \sqrt{12}$). $\bar{r}_{t+j \rightarrow t+h}$ indicates the portfolio is implemented j years after the *Dur* measurement and is held for $h - j$ years. Moreover, $\bar{r}_{t \rightarrow t+1}^{Large}$ indicates the portfolio is based on firms with market equity above the 80% NYSE quantile, and $\bar{r}_{t \rightarrow t+1}^{dist adj}$ indicates returns are adjusted for delistings. Panel B reports β s and annualized α s ($\times 12$) from factor regressions. Statistical inference is robust to heteroskedasticity and autocorrelation (Newey and West, 1987; Newey and West, 1994) with t_{stat} in parentheses and p -values in brackets.

Panel A: Average returns and Sharpe ratios													
Duration decile	Value-weighted portfolios						Duration decile	Equal-weighted portfolios					
	$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	$\bar{r}_{t \rightarrow t+1}^{dist adj}$	$\bar{r}/\sigma$		$\bar{r}_{t \rightarrow t+1}$	$\bar{r}_{t \rightarrow t+5}$	$\bar{r}_{t+4 \rightarrow t+5}$	$\bar{r}_{t \rightarrow t+1}^{Large}$	$\bar{r}_{t \rightarrow t+1}^{dist adj}$	$\bar{r}/\sigma$
Short	12.2%	11.3%	10.0%	9.8%	12.0%	0.61	Short	12.9%	12.5%	11.4%	11.2%	12.8%	0.60
2	10.4%	9.9%	9.1%	8.5%	10.4%	0.60	2	12.4%	12.1%	12.2%	9.6%	12.4%	0.63
3	10.3%	9.3%	10.1%	8.3%	10.4%	0.58	3	11.8%	12.1%	12.5%	9.7%	11.8%	0.61
4	10.4%	9.6%	8.8%	7.7%	10.4%	0.62	4	11.7%	11.9%	11.0%	9.2%	11.7%	0.62
5	8.6%	9.3%	9.4%	7.9%	8.6%	0.51	5	11.5%	11.5%	11.8%	8.9%	11.5%	0.61
6	9.3%	8.7%	8.8%	7.8%	9.4%	0.59	6	11.1%	11.4%	11.2%	8.9%	11.1%	0.57
7	9.1%	9.0%	9.2%	6.9%	9.1%	0.54	7	10.3%	10.7%	10.3%	7.8%	10.3%	0.53
8	7.8%	7.5%	7.2%	6.3%	7.8%	0.49	8	9.3%	9.6%	10.0%	6.9%	9.2%	0.45
9	7.2%	6.7%	7.6%	6.0%	7.3%	0.44	9	7.5%	9.0%	10.6%	6.9%	7.5%	0.33
Long	6.0%	5.9%	8.2%	5.4%	6.1%	0.29	Long	5.0%	7.7%	9.3%	5.1%	4.8%	0.18
L-S (t_{L-S})	−6.1% (−1.93)	−5.4% (−2.02)	−1.8% (−0.59)	−4.4% (−1.31)	−6.0% (−1.89)	−0.36 [0.02]	L-S (t_{L-S})	−7.8% (−2.74)	−4.8% (−1.96)	−2.1% (−0.77)	−6.1% (−2.15)	−8.0% (−2.76)	−0.48 [0.00]
Panel B: Risk-adjusted performance based on factor models													
Duration decile	CAPM			Fama and French (2015) 5-factors					Hou, Xue, and Zhang (2015) q-factors				
	α_{CAPM}	β_{MKT}	α_{FF}	β_{MKT}	β_{SMB}	β_{HML}	β_{CMA}	β_{RMW}	α_q	β_{MKT}	β_{SIZE}	β_{INV}	β_{ROE}
Value-weighted portfolios													
Short	4.0%	1.03	0.0%	1.08	0.44	0.66	−0.12	0.09	2.7%	1.04	0.32	0.53	−0.29
2	3.3%	0.95	−0.8%	1.04	0.21	0.45	0.04	0.27	0.6%	1.00	0.12	0.49	−0.02
3	3.0%	0.99	−1.1%	1.07	0.29	0.41	0.04	0.28	0.5%	1.04	0.17	0.48	−0.05
4	3.4%	0.96	−0.2%	1.04	0.17	0.26	0.19	0.24	1.4%	1.01	0.07	0.43	−0.04
5	1.9%	0.93	−1.5%	1.02	0.12	0.19	0.20	0.29	−0.2%	0.99	0.03	0.37	0.04
6	2.8%	0.91	−0.3%	1.00	0.03	−0.01	0.34	0.33	−0.2%	0.98	0.00	0.33	0.20
7	2.1%	0.98	−0.8%	1.06	0.08	−0.02	0.22	0.39	−0.5%	1.03	0.02	0.23	0.21
8	0.7%	0.94	−0.6%	0.98	0.02	−0.17	0.28	0.19	−0.6%	0.97	0.02	0.05	0.18
9	0.2%	0.98	−0.5%	1.01	−0.06	−0.33	0.19	0.33	−0.6%	1.00	−0.07	−0.17	0.28
Long	−2.6%	1.21	0.4%	1.10	0.04	−0.46	−0.15	−0.01	0.8%	1.12	0.00	−0.63	0.05
L-S (t_{L-S})	−6.6% (−1.88)	0.18 (1.73)	0.4% (0.16)	0.02 (0.34)	−0.40 (−5.53)	−1.12 (−10.41)	−0.03 (−0.15)	−0.10 (−0.95)	−2.0% (−0.64)	0.08 (1.21)	−0.32 (−2.92)	−1.15 (−5.69)	0.34 (2.02)
Equal-weighted portfolios													
Short	4.3%	1.12	−1.6%	1.15	0.80	0.67	−0.03	0.23	1.3%	1.10	0.64	0.68	−0.29
2	4.3%	1.08	−1.0%	1.10	0.72	0.45	0.07	0.27	1.0%	1.06	0.60	0.53	−0.12
3	3.8%	1.07	−1.5%	1.11	0.68	0.37	0.13	0.32	0.5%	1.06	0.55	0.48	−0.07
4	3.9%	1.06	−0.6%	1.08	0.63	0.27	0.09	0.33	0.7%	1.05	0.52	0.37	0.01
5	3.5%	1.08	−0.3%	1.08	0.59	0.21	0.04	0.30	0.9%	1.05	0.49	0.25	0.02
6	2.9%	1.12	−0.1%	1.10	0.59	0.08	0.08	0.23	1.1%	1.07	0.50	0.13	0.00
7	2.0%	1.13	0.6%	1.06	0.55	0.00	−0.02	0.07	2.0%	1.04	0.48	−0.10	−0.08
8	0.5%	1.19	0.2%	1.09	0.52	−0.13	−0.12	0.07	1.4%	1.08	0.44	−0.30	−0.04
9	−2.3%	1.33	0.2%	1.14	0.51	−0.38	−0.13	−0.21	1.2%	1.16	0.46	−0.60	−0.17
Long	−6.2%	1.53	−3.0%	1.27	0.75	−0.33	−0.12	−0.55	−0.5%	1.30	0.64	−0.53	−0.58
L-S (t_{L-S})	−10.5% (−3.26)	0.41 (3.64)	−1.4% (−0.90)	0.12 (2.99)	−0.06 (−0.97)	−1.00 (−12.25)	−0.09 (−0.64)	−0.78 (−5.68)	−1.8% (−0.66)	0.19 (2.29)	0.00 (0.02)	−1.21 (−5.15)	−0.28 (−1.14)

absence of mispricing, while mispricing generates an even larger short duration premium among small firms.

6. Conclusion

In this paper, I develop a new measure of equity cash flow duration and use it to empirically study the premium for stocks with cash flows concentrated in the short term

(i.e., the short duration premium). I find that several of the short duration premium properties (including its time variation) are consistent with the idea that reinvestment risk, defined as exposure to declines in expected wealth growth and priced by long-term investors, is its main driver. Moreover, I show that the value and profitability premia can be explained by the lower cash flow duration of value and profitable companies.

My results are largely consistent with (and strengthen the message of) the equity term structure literature (reviewed in Binsbergen and Koijen, 2017). New models attempting to explain the equity term structure also need to generate its connection to the cross-section of stock return. In particular, they need to produce a short duration premium and its link to the value and profitability premia. The ICAPM I explore in this paper is one such model (as Gonçalves, 2021 shows that it can also explain the equity term structure), but it does not provide a microfoundation for the time-varying expected returns that generate reinvestment risk. Consequently, exploring new models that microfound time-varying expected returns and are consistent with the equity term structure and short duration premium is a fruitful path for future research. Providing other frameworks that can also jointly explain the equity term structure and short duration premium would be of great value to the literature as well.

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